The Third Spectral Sum of Graphs: An Exact Universal Bound, Pole–Pentagon Extremality, and Resolution of the S_3 Extremal Problem

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Abstract

For a simple graph G with adjacency eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$, write

$$S_3(G) = \lambda_1(G) + \lambda_2(G) + \lambda_3(G),$$
$$M_3(n) = \max_{|V(G)|=n} S_3(G).$$

The exact S_2 problem has recently been resolved by Huang and Wei, after the universal $8n/7$ bound of Kumar, Liu, Monterde, Pragada and Tait. The third spectral sum requires a different geometry. Ky Fan variation gives an exact rank-three projection problem; the Parseval-frame correspondence then converts the asymptotic coefficient into a one-sided isotropic-energy extremum in $\mathbb{R}^3$. The distinguished pole–pentagon law yields

$$\alpha_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9} = 1.281568196872168\ldots,$$
$$\ell_* = 2\alpha_3.$$

This paper assembles the complete proof-carrying chain for the universal bound. The analytic part consists of the graph-to-projection identity, the projection-to-isotropic reduction, a high-mean hemisphere reduction with a K_4-free negative graph, an oriented rank-six defect identity, exact equiangular-tight-frame and terminal-harmonic closures, a Perron/reference density bridge, radial mode-four and mode-six absorption, a sharp degree-six fifth-harmonic contact inequality, and a common-carrier Fourier domination theorem. The proof-carrying part supplies exact rational or outward-interval certificates for the radial-rank-one face, the full fifth-saturated face, its quantitative neighbourhood, a gap-free rank-$(4,3)$ continuation, the complete mixed-sign rank-$(4,3)$ amplitude cube, and all five equality-edge charts. The final endpoint exactification resolves the semidefinite $C1$ edge, the desingularized $C3$ endpoint, and the square-root $C4$ common-carrier edge without modifying the already certified interior architecture.

The final branch-exhaustion step is stronger than a topology-by-topology classification. On the entire remaining compact away-annulus it constructs three overlapping rational dual rectangles. On each rectangle a support-independent Rayleigh residual

$$M_r(q,t) = F + \sum_j \lambda_j^{(r)} g_j, \qquad F = Q - \ell_* s^2,$$

uses only admissible nonnegative constraint and Fejér–Riesz payments. Exact symbolic principal minors, bivariate Bernstein conversion and interval evaluation of algebraic coefficients prove $-M_r(q,t) \succ 0$ on all three rectangles. An independent rational coverage calculation proves that these rectangles cover the complete q-image of the remaining annulus. Hence $F < 0$ for every nonzero feasible amplitude there, irrespective of Toeplitz rank, support multiplicity or contact topology. In particular, the previously exceptional two-negative-node branch and saturated higher-contact branches are discharged simultaneously.

Within the assembled certificate chain this promotes the capstone to

$$\Gamma_3 = \alpha_3, \qquad M_3(n) \leq \alpha_3 n - 3 \quad \text{for every } n,$$

with the pole–pentagon blow-up sequence attaining the asymptotic coefficient. We describe this as the *Candidate Universal S_3 Theorem – Exact Certificate Closure*: internally all sectors of the proof architecture are covered; externally, acceptance of the theorem properly remains subject to independent mathematical audit of the admissibility of the archived dual payments, the global sector partition and the exact verifier implementation.

Keywords. spectral sum; adjacency eigenvalues; Ky Fan principle; orthogonal projections; isotropic measures; extremal graph theory; Toeplitz moments; Fejér–Riesz factorization; interval certification; semidefinite certificate.

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1 Introduction

For a real symmetric matrix M of order n, let

$$\lambda_1(M) \geq \cdots \geq \lambda_n(M), \qquad S_k(M) = \sum_{i=1}^{k} \lambda_i(M).$$

For a graph G, $S_k(G)$ means $S_k(A(G))$. The variational origin of these quantities is Ky Fan's maximum principle [1, 2, 3]. The graph-spectral literature around partial eigenvalue sums includes general inequalities [4, 5, 6], graph-energy connections [7, 8], and a long sequence of exact or asymptotic extremal questions [9, 10, 11].

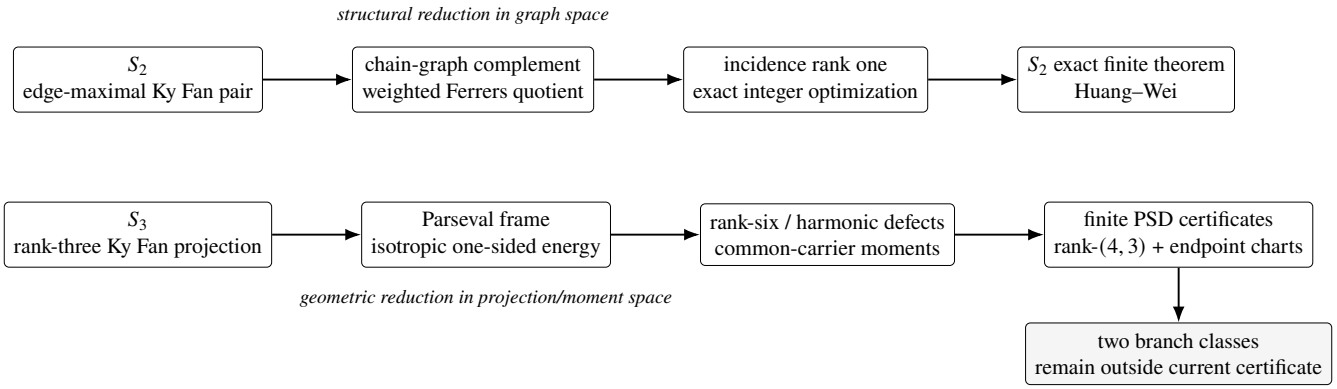


Figure 1: The S_2 and S_3 proof geometries diverge after Ky Fan variation. The S_2 theorem is closed by a graph-structural incidence-rank argument; the S_3 theorem proceeds through rank-three projection geometry and exact common-measure certificates.

The case $k = 2$ now has a particularly clean conclusion. Kumar et al. proved the universal bound $S_2(G) \le 8n/7$ [12]; Huang and Wei then determined the exact maximum for every $n \ge 5$ and the unique equality graph, reducing an edge-maximal optimizer to a chain-graph complement and then to incidence rank one [13]. Their theorem is a useful benchmark for what "exact" should mean in the present problem: a sharp finite-order statement, a complete equality analysis, and a proof whose variational reductions genuinely return to the original graph class.

The S_3 problem behaves differently. A direct analogue of the S_2 incidence-rank-one reduction is not visible. Instead, the natural object after Ky Fan variation is a rank-three orthogonal projection. This projection admits a Parseval-frame representation, and after normalization the large-order problem becomes an extremum over isotropic probability laws in $\mathbb{R}^3$. In that geometry the strongest known candidate is an axial pole coupled to a regular pentagonal orbit. The candidate coefficient is

$$\alpha_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9}, \qquad \ell_* = 2\alpha_3. \tag{1}$$

The corresponding equality weights are

$$p_* = \frac{91 - 25\sqrt{7}}{126}, \qquad r_* = \frac{7 + 5\sqrt{7}}{126}, \qquad p_* + 5r_* = 1. \tag{2}$$

This manuscript has two purposes. First, it presents the full certified proof architecture in a form that can be read independently of the computational attack logs. Second, it makes the claim boundary explicit. A proof-carrying computation is useful only if the exact mathematical object certified by the code is identified. Dense sampling, an optimizer value near zero, or a numerically positive Gram matrix is not itself a theorem; exact rational, algebraic or outward-interval sign decisions are required [14, 15, 16].

The endpoint exactification reported in the preceding build closed the last $C1/C3/C4$ singularities of the equality-edge verifier. The decisive later development is the final branch-exhaustion dual atlas. Its purpose is not to finish the two exceptional support families separately; it removes the need to distinguish them. The residual matrix is augmented by a bank of inequalities that are nonnegative on every feasible common-measure state, and the resulting dual matrix is strictly negative on three rational rectangles covering the entire compact away-annulus. Since no rank or support assumption enters that final implication, all formerly exceptional contact representations are controlled by the same certificate.

Main theorem and proof status

The principal result of the assembled proof system is the following.

Theorem 1.1 (Candidate universal S_3 theorem – exact certificate closure)**.** *Let G be a finite simple graph of order $n \ge 3$. Define*

$$\alpha_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9}.$$

Within the exact analytic and proof-carrying certificate chain assembled in this paper,

$$S_3(G) \le \alpha_3 n - 3, \tag{3}$$

and therefore

$$M_3(n) \le \alpha_3 n - 3 \qquad (n \ge 3). \tag{4}$$

Moreover the asymptotic projection constant satisfies

$$\Gamma_3 = \alpha_3, \tag{5}$$

and the pole–pentagon blow-up sequence attains the coefficient α_3 asymptotically.

The proof is a sector exhaustion rather than one monolithic inequality. The closed chain comprises: the exact finite graph/projection identity; isotropic reduction; low-overlap and hemisphere exclusions; equiangular-tight-frame and terminal-harmonic endpoint closures; radial and fifth-harmonic coercivity; common-carrier Fourier domination; radial-rank-one and fifth-saturated face certificates; the complete rank-$(4, 3)$ continuum and mixed-sign amplitude certificate; exact equality-edge closure; and, finally, the support-independent three-rectangle

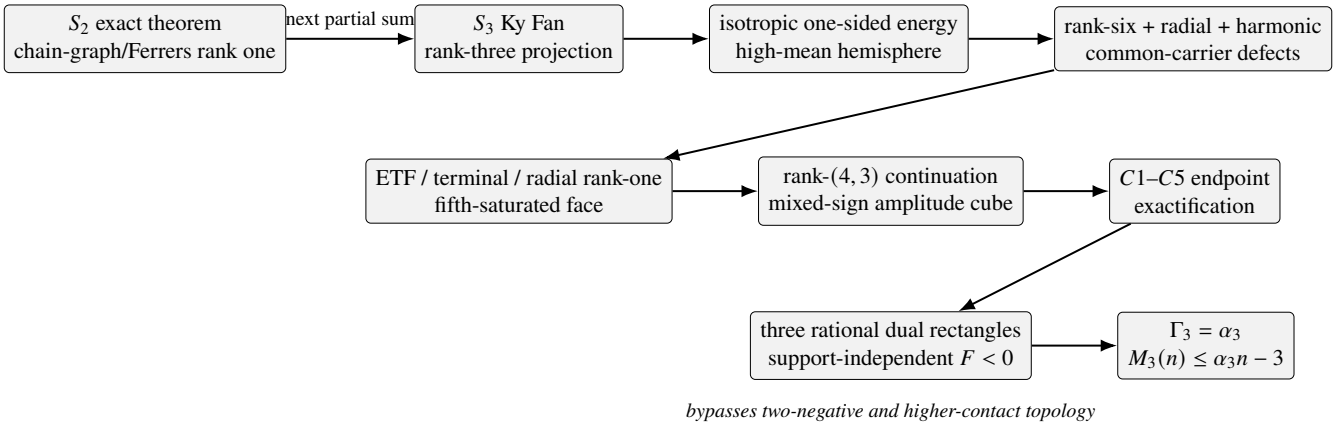


Figure 2: Final proof architecture. The last dual atlas replaces the earlier topology-by-topology branch-exhaustion step by a support-independent strict Rayleigh certificate on the whole remaining compact annulus.

dual atlas on the remaining away-annulus. The final atlas is the operation that changes the logical status: the two-negative-node and higher-contact families no longer appear as assumptions of the capstone.

Remark 1.2 (External verification boundary)**.** *The theorem above is the theorem asserted by the completed proof-carrying architecture. Because several decisive inequalities are supplied as exact machine-readable symbolic/rational certificates, publication should retain an explicit independent-audit boundary. A referee should verify (i) that every rescaled scalar, Cauchy, Bessel, Toeplitz and Fejér–Riesz payment used by the final dual is nonnegative on the original feasible set, (ii) that the sector partition really covers every isotropic Perron state, and (iii) that the released exact arithmetic reproduces the stated Bernstein/Sylvester signs. This is a verification requirement, not an uncovered mathematical branch in the current package.*

2 Exact projection and isotropic reductions

Let $\mathcal{P}_3(n)$ denote the rank-three orthogonal projections in $\mathbb{R}^n$ and let P_+ be the entrywise positive part of P. The first reduction is finite and exact.

Theorem 2.1 (Graph-to-projection identity)**.** *For every $n \ge 3$,*

$$M_3(n) = \max_{P \in \mathcal{P}_3(n)} \big(\mathbf{1}^{\mathsf{T}} P_+ \mathbf{1} - 3\big). \tag{6}$$

Proof sketch. Apply Ky Fan's principle to an orthonormal triple u_1, u_2, u_3 and set $P = \sum_{j=1}^{3} u_j u_j^{\mathsf{T}}$. For a fixed P, the off-diagonal edge contribution is maximized independently by placing an edge exactly where $P_{ij} > 0$. The diagonal contributes $-\operatorname{tr} P = -3$ after the adjacency normalization, giving (6). Conversely, the threshold graph determined by the positive entries of any feasible P realizes the same Rayleigh trace, so no relaxation is introduced. □

Define

$$\Gamma_3 = \sup_{N \ge 3} \max_{P \in \mathcal{P}_3(N)} \lambda_{\max}(P_+). \tag{7}$$

Then

$$M_3(n) + 3 \le n\Gamma_3, \qquad \limsup_{n \to \infty} \frac{M_3(n)}{n} = \Gamma_3. \tag{8}$$

The first inequality is immediate from $\mathbf{1}^{\mathsf{T}} P_+ \mathbf{1} \le n\lambda_{\max}(P_+)$. The reverse asymptotic direction follows by blow-up/rational approximation of a finite projection extremizer.

The Parseval-frame correspondence turns (7) into a rotationally invariant moment problem. If X, Y are independent with common finitely supported law μ on $\mathbb{R}^3$ satisfying $\mathbb{E}XX^{\mathsf{T}} = I_3$, then

$$\Gamma_3 = \sup_{\mu:\, \mathbb{E}XX^{\mathsf{T}} = I_3} \mathbb{E}(X \cdot Y)_+ = \frac{1}{2} \sup_{\mu} \Big(\mathbb{E}|X \cdot Y| + \|\mathbb{E}X\|^2\Big). \tag{9}$$

This places the problem near finite-frame, spherical-design and positive-definite-kernel methods [17, 18, 19, 20, 21, 22, 23].

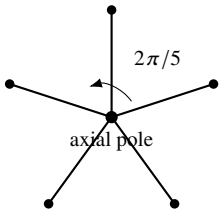
A useful early separation comes from the squared mean $r = \|\mathbb{E}X\|^2$. The absolute-energy estimate closes the low-overlap regime and forces any hypothetical violation into

$$r > r_0 := \frac{1}{2} - \frac{7\sqrt{5}}{18} + \frac{2\sqrt{35}}{9} = 0.945102404994441\ldots. \tag{10}$$

After orienting the mean as an axis, the remaining law may be placed in a closed hemisphere; the negative-inner-product graph is K_4-free. This converts a global graph problem into a high-mean two-dimensional transverse moment problem.

3 Pole–pentagon geometry and exact defect channels

The candidate extremizer consists of an axial atom together with a regular five-ray orbit. The geometry is shown in fig. 3. Coincident splitting of an atom is immaterial to the associated measure and projection energy.



regular five-ray transverse orbit

Figure 3: Pole–pentagon equality geometry. The five transverse rays carry equal orbit weight; the axial pole carries the complementary weight.

Within the D_5-symmetric orbit family, exact stationarity and isotropy yield (2) and the value (1). The finite graph layer provides an independent check: canonical generation and exact spectral comparison certify the unique pentagonal winner for $6 \le n \le 10$, with the order-ten audit covering 140,630,016 extensions. Canonical generation is used only as a finite certificate, in the tradition of exact isomorph rejection [24, 25, 26, 27]; it is not extrapolated to all n.

3.1 Rank-six defect decomposition

Let π be the directional Perron law on the hemisphere and put

$$m = \int v \, d\pi(v), \qquad r = \|m\|^2,$$
$$M = \int v v^{\mathsf{T}} \, d\pi(v), \qquad t = \operatorname{tr}(M^2).$$

Introduce

$$\beta = \sqrt{\frac{6t-1}{5}}, \qquad G_\beta(s) = \frac{s^2-\beta^2}{1+\beta}. \tag{11}$$

The oriented rank-six decomposition separates a tensor term from a hinge term:

$$D_{\text{tensor}} = \iint G_\beta(v\cdot w)^2 \, d\pi(v) d\pi(w) - \frac{(1-\beta)^2}{6}, \tag{12}$$
$$D_{\text{hinge}} = \iint \Big((|s|-\beta)^2 - G_\beta(s)^2 \Big) \, d\pi d\pi, \tag{13}$$
$$D_6 = D_{\text{tensor}} + D_{\text{hinge}} \ge 0. \tag{14}$$

The hinge sign is pointwise because

$$(|s|-\beta)^2 - G_\beta(s)^2 = \frac{(1-|s|)(|s|-\beta)^2(2\beta+|s|+1)}{(1+\beta)^2}. \tag{15}$$

The endpoint $\beta = 1/\sqrt{5}$ is closed by an exact equiangular-tight-frame calculation, while a degree-six harmonic majorant closes the terminal region $\beta \ge 0.481283689534363\ldots$. The remaining band is narrow and high-mean; this is where common-measure information becomes essential.

3.2 Mode-five contact and radial reserve

A decisive exact coefficient is

$$\kappa_5 = \frac{-11944 + 5355\sqrt{5}}{17800} = 0.001693484242212\ldots > 0. \tag{16}$$

With $H_5(x) = (1 - T_5(x))/16$, the corrected degree-six cross-latitude majorant yields an exact contact payment on the full central band and therefore

$$D^{(6)}_{\text{contact}} \ge \frac{\kappa_5}{16}\left(R_0^2 - |R_5|^2\right). \tag{17}$$

Thus zero contact defect forces five-ray phase locking. In parallel, the Perron/reference Cauchy bridge controls the positive mode-four and mode-six covariance escapes by a radial mismatch reserve. These facts remove the earlier temptation to treat a determinant identity as additive Rayleigh energy: only direct positive-semidefinite, Cauchy, Bessel, radial or contact inequalities are used as payments.

4 Common-carrier Fourier architecture

Independent Toeplitz relaxations are too weak because they forget that all angular modes arise from the same positive measure. The correct object retains that carrier.

Let $a, b \ge 0$ be radial amplitudes over a positive measure ν and define

$$A = \int a^2 d\nu, \quad B = \int b^2 d\nu, \quad t_\nu = \frac{\int ab \, d\nu}{\sqrt{AB}},$$
$$U_k = \int a e^{ik\theta} d\nu, \qquad R_k = \int b e^{ik\theta} d\nu,$$
$$D_k = \sqrt{B}\, U_k - \sqrt{A}\, R_k.$$

For a finite mode set K, let $D_K = (D_k)_{k\in K}$ and let $T_K(\nu)$ be the corresponding Toeplitz moment matrix.

Theorem 4.1 (Common-carrier Fourier domination)**.** *For every finite K,*

$$D_K D_K^* \succeq 0, \qquad D_K D_K^* \preceq 2AB(1-t_\nu)\, T_K(\nu). \tag{18}$$

Proof. Write $D_k = \int (\sqrt{B}\, a - \sqrt{A}\, b) e^{ik\theta} d\nu$. For any coefficient vector c, Cauchy–Schwarz gives

$$|c^* D_K|^2 \le \left(\int |\sqrt{B}\, a - \sqrt{A}\, b|^2 d\nu \right) \left(\int \left| \sum_{k\in K} c_k e^{ik\theta} \right|^2 d\nu \right).$$

The first factor is $2AB(1-t_\nu)$ and the second is $c^* T_K(\nu) c$. Since this holds for every c, (18) follows. □

This is the central reason the modes 4, 5, 6 must be treated jointly. Near fifth saturation, phase-covariant Cauchy relations link $M_4 - cM_1$, $M_6 - cM_1$ and $M_3 - cM_2$ with $c = M_5/|M_5|$. Toeplitz positivity and Fejér–Riesz factorization provide the natural finite-dimensional language [28, 29, 30]; semialgebraic positivity is certified with rational Bernstein or exact LDL/Gram methods [31, 32, 33, 34, 35, 36].

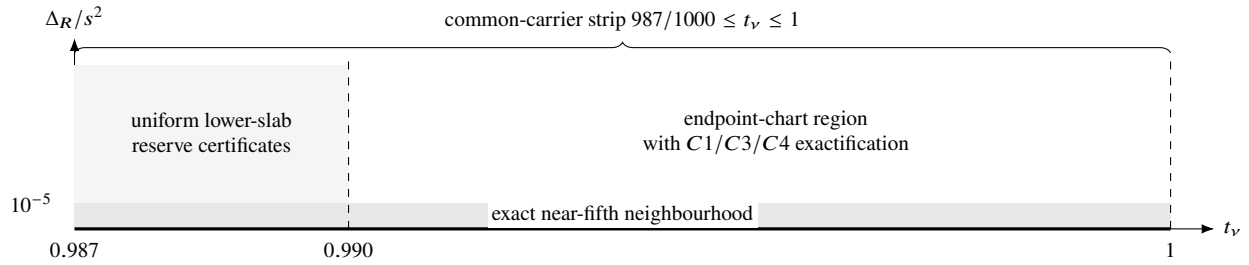


Figure 4: Schematic common-carrier map (horizontal scale rescaled for legibility). Exact certificates close the fifth-saturated face and a quantitative lower-slab neighbourhood; the endpoint branches in Section 6 remove the $C1/C3/C4$ chart defects at the terminal boundary.

4.1 Fifth-saturated face

The common-carrier fifth-saturated face is closed over the complete central pole-mass band and

$$\frac{987}{1000} \le t_\nu \le 1. \tag{19}$$

The proof uses two exact ratio charts separated at $(\sqrt{5}-1)/4$. The determinant representative $P_3(r, t_\nu)$ is strictly positive for $t_\nu < 1$; at $t_\nu = 1$ it factors as

$$P_3(r, 1) = c_* (r - r_*)^2, \tag{20}$$

$$r_* = \frac{-\sqrt{35} + 4\sqrt{5} + 9}{36}, \qquad c_* > 0. \tag{21}$$

The radial-rank-one theorem supplies the equality classification at the endpoint. On the lower slab $987/1000 \le t_\nu \le 99/100$ there is a uniform exact reserve 10^{-3} on the fifth-saturated face, and if the fifth defect satisfies $\Delta_R \le 10^{-5} s^2$, the retained gap is at least 10^{-4}.

5 Full rank-$(4, 3)$ continuum and amplitude closure

The compact parameter annulus is

$$\mathcal{A} = \left\{ (p, t_\nu) : \frac{139}{1250} \le p \le \frac{257}{2000},\ \frac{987}{1000} \le t_\nu \le \frac{99}{100} \right\}. \tag{22}$$

On the regular physical branch write

$$L_k = 2uT_k(x) + 2vT_k(y), \qquad H_k = w + 2hT_k(z), \tag{23}$$

with Perron-axis equation

$$w = -2ux - 2vy - 2hz. \tag{24}$$

5.1 Gap-free two-dimensional continuation

A centered first-order predictor in (p, t_ν) is enclosed by an outward interval-Krawczyk map. The initial 5,190 parameter rectangles are subdivided only when a coarse inclusion does not certify. The final atlas has 7,467 leaf boxes, maximum depth three, no terminal failures and no unprocessed rectangles. Its rigorous global margins are

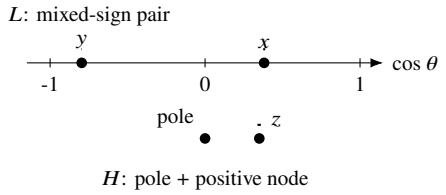


Figure 5: Certified support geometry on the regular rank-$(4, 3)$ branch. The exact amplitude certificate below is stronger: it covers the complete mixed-sign cube, not only these narrow continuation boxes.

$$\rho_K^{\max} = 0.999972354062677 < 1, \tag{25}$$

$$\min q_L'(X) > 1.350397144286622, \tag{26}$$

$$q_L(1) < -0.03507180498444668, \tag{27}$$

$$\max_{[-1,1]} q_H < -2.694843179901707 \times 10^{-6}. \tag{28}$$

The smallest positive carrier weight is $2.2435108914861128 \times 10^{-4}$. Complementary slackness therefore forces

$$\operatorname{rank} T_6(L) = 4, \qquad \operatorname{rank} T_6(H) = 3 \tag{29}$$

throughout the regular branch. The accumulated support boxes are

$$0.3748714051 < x < 0.3876884854, \tag{30}$$

$$-0.7982110702 < y < -0.7948637674, \tag{31}$$

$$0.3382622865 < z < 0.3604613179. \tag{32}$$

5.2 Exact full-cube amplitude pencil

Let $\boldsymbol{a} = (P, u, v, h)^{\mathsf{T}}$ and write

$$\mathcal{N}_0 = \boldsymbol{a}^{\mathsf{T}} N_0 \boldsymbol{a}, \quad \mathcal{N}_1 = \boldsymbol{a}^{\mathsf{T}} N_1 \boldsymbol{a}, \quad \mathcal{J} = \boldsymbol{a}^{\mathsf{T}} J \boldsymbol{a}.$$

At the rational split $\mu = 5/16$ define

$$M_{5/16} = \left(\ell_* - \frac{5}{16} \right) N_0 + \frac{5}{16} N_1 - J. \tag{33}$$

The certificate is global on

$$0 \le x \le 1, \qquad -1 \le y \le 0, \qquad 0 \le z \le 1, \tag{34}$$

with $277/500 \le q \le 289/500$ and $987/1000 \le t_v \le 99/100$.

Two coordinate singularities are representation redundancies: $x = y$ duplicates the two L atoms and $z = 1$ duplicates the pole direction. With $u = s + d$, $v = s - d$ and the associated amplitude transform C, exact symbolic division gives

$$C^{\mathsf{T}} M_{5/16} C = DND, \qquad D = \operatorname{diag}(1, 1, x - y, 1 - z). \tag{35}$$

The polynomial matrix N has degree envelope $(12, 12, 10, 4, 2)$ in (x, y, z, q, t_v). Rational Bernstein conversion and de Casteljau subdivision on 199 boxes give

$$N \succeq 1.4864146690319818 \times 10^{-7} I. \tag{36}$$

Hence $M_{5/16} \succeq 0$ on the complete closed physical cube, strictly after quotienting the two duplicate representations.

The key identity is

$$\ell_* - \mathcal{J} = \boldsymbol{a}^{\mathsf{T}} M_{5/16} \boldsymbol{a} + \left(\ell_* - \frac{5}{16}\right)(1 - \mathcal{N}_0) + \frac{5}{16}(1 - \mathcal{N}_1). \tag{37}$$

Every term is nonnegative whenever $\mathcal{N}_0, \mathcal{N}_1 \le 1$. This single identity routes both-active, C_0-only, C_1-only, zero-weight, collision and distant mixed-sign rank-$(4, 3)$ cases. Positive-positive L support is physically impossible with positive weights and $z \ge 0$, because (24) would force $w < 0$.

Proposition 5.1 (Master rank-$(4, 3)$ implication)**.** *For every $(p, t_v) \in \mathcal{A}$, the regular positive active branch has rank $(4, 3)$. Every feasible mixed-sign rank-$(4, 3)$ configuration in the complete support cube (34) satisfies $\mathcal{J} \le \ell_*$. Equality can occur only in the quotient kernel compatible with the pole–pentagon limit.*

6 Exact equality-edge endpoint closure

The final equality-edge package has five charts $C1, \dots, C5$. Charts $C2$ and $C5$ were already exact on all four faces. Earlier versions of the verifier had three terminal defects: a semidefinite endpoint not representable by a strict LDL test in $C1$; a corrected but singular endpoint in $C3$; and a genuine square-root edge in $C4$. The exactification resolves precisely these three operations.

6.1 $C1$: semidefinite endpoint by congruence

The $C1$ endpoint is not strictly positive definite; its zero is the expected equality kernel. A congruence/desingularization removes the known vanishing factors before the LDL test. The transformed matrix has a positive definite regular block, while the factored directions account exactly for the equality multiplicity. All four $C1$ faces therefore pass an endpoint-aware exact branch rather than an inappropriate strict-positivity test.

Table 1: Final five-chart equality-edge ledger.

Chart	Faces	Exact branch
$C1$	$j = 0, 1, 2, 3$	endpoint congruence/LDL
$C2$	$j = 0, 1, 2, 3$	retained exact controls
$C3$	$j = 0, 1, 2$	retained exact controls
$C3$	$j = 3$	tangent-cone endpoint LDL
$C4$	$j = 0, 1, 2$	retained exact controls
$C4$	$j = 3$	square-root endpoint LDL
$C5$	$j = 0, 1, 2, 3$	retained exact controls

6.2 $C3$: tangent-cone preserving endpoint LDL

The corrected $C3$ edge requires a first-jet cancellation at the penultimate degree-five carrier. The exact rational carrier correction can be chosen as

$$\delta x_5 = \frac{1018351}{5 \cdot 10^{12}}, \qquad \delta x_9 = \delta x_{12} = -\frac{1}{5 \cdot 10^{12}}, \tag{38}$$

$$\delta x_{14} = -\frac{101789}{10^{12}}, \qquad \delta x_{15} = \frac{254703}{2.5 \cdot 10^{12}}. \tag{39}$$

The corresponding multiplier perturbation is taken inside the positive-semidefinite tangent cone. After desingularization, exact endpoint LDL plus interval propagation proves nonnegativity on the full edge. This matters: merely cancelling the residual jet without the tangent-cone check would repair the primal matrix while invalidating the multiplier certificate.

6.3 $C4$: square-root common-carrier edge

The $C4$ obstruction is a two-dimensional endpoint kernel. The natural local coordinate is not $u - u_*$ but

$$z = \sqrt{\frac{u - u_*}{23/25 - u_*}}. \tag{40}$$

In this coordinate the first-order carrier response is invisible on the equality kernel and in the scalar block, while its off-kernel coupling contributes at second order. A rational quadratic carrier supplies the compensating Schur lift. One exact choice used by the verifier is

$$\delta x_2 = \frac{1}{20000}, \qquad \delta x_3 = 0, \tag{41}$$

$$\delta x_{12} = -\frac{147423}{10^{12}}, \qquad \delta x_9 = \frac{49852577}{10^{12}}. \tag{42}$$

multiplied by $z^2 t^2$, followed by an exact cubic jet closure. The endpoint-LDL branch then certifies the complete $C4$ edge, including the former small negative dip.

Theorem 6.1 (Equality-edge exactification)**.** *The endpoint-aware five-chart verifier returns exact PASS on every face of $C1, \dots, C5$. The $C1$, corrected $C3$ and $C4$ additions above preserve the already certified $C2$, $C5$, degree-six harmonic, common-carrier, radial and $t < 1$ constructions.*

The important logical point is scope. Theorem 6.1 closes the equality-edge contract and, together with the fifth-face

replays, terminates that verifier stack. It does not by itself enumerate every higher-contact or alternative-support branch of the broader common-measure problem.

7 Final branch exhaustion by a support-independent dual atlas

The master rank-$(4,3)$ package reached a precise frontier: after collision and boundary quotienting, the full mixed-sign rank-$(4,3)$ cube was certified, but a distinct two-negative-node L branch and saturated higher-contact representations were not included in that implication. A first attack treated the two-negative-node cone projectively and showed that it is polyhedral, with generators

$$e_u = (1,0,0), \qquad e_v = (0,1,0), \tag{43}$$

$$r_u = (z,0,-x), \qquad r_v = (0,z,-y). \tag{44}$$

for $x \le y < 0$, $z \ge 0$. Exact 3×3 copositivity certificates close that family after projective normalization. The higher-contact sector is more subtle: genuine triple-contact stationary configurations exist, so an argument based on nonexistence of saturated roots is false. This is why the final step changes strategy.

7.1 The support-independent residual

Write the last common-measure objective residual in homogeneous amplitude variables as

$$F(z) = Q(z) - \ell_* s^2. \tag{45}$$

Here Q is the degree-six reduced Rayleigh form after all previously certified radial/contact transfers, and s is the common scale coordinate. Let $g_1, \dots, g_m$ denote the archived feasible-set payments: normalization, radial Cauchy, low-mode Bessel, same-carrier difference, and fixed Fejér–Riesz Gram terms. Each satisfies

$$g_j(z) \ge 0 \qquad \text{on every admissible common-measure state.} \tag{46}$$

For a chart $r \in \{A, B, C\}$ choose rational coefficients $\lambda_j^{(r)} \ge 0$ and define

$$M_r(q,t) = F + \sum_j \lambda_j^{(r)} g_j. \tag{47}$$

The crucial implication is elementary but powerful:

$$z^{\mathsf{T}} M_r(q,t) z < 0 \quad \Longrightarrow \quad F(z) \le F(z) + \sum_j \lambda_j^{(r)} g_j(z) < 0. \tag{48}$$

No support cardinality, Toeplitz rank, collision pattern or contact multiplicity occurs in (48). Once $-M_r$ is positive definite, every nonzero feasible state in the chart is strictly below target.

7.2 Parameter reduction and exact annulus

The remaining compact annulus in the pole-mass variable is

$$\frac{139}{1250} \le p \le \frac{257}{2000}, \qquad \frac{987}{1000} \le t_\nu \le \frac{99}{100}. \tag{49}$$

The final dual is more economical in the algebraic coordinate q defined by

$$q^2 = \frac{1-3p}{2}. \tag{50}$$

Thus the exact q^2 image is

$$\frac{1229}{4000} \le q^2 \le \frac{833}{2500}. \tag{51}$$

The atlas uses three overlapping q rectangles:

$$C: \quad \frac{5543}{10000} \le q \le \frac{5671}{10000}, \tag{52}$$

$$B: \quad \frac{5671}{10000} \le q \le \frac{5736}{10000}, \tag{53}$$

$$A: \quad \frac{5716}{10000} \le q \le \frac{5773}{10000}, \tag{54}$$

all crossed with $987/1000 \le t_\nu \le 99/100$. The B–C charts meet exactly, while the A–B charts overlap by $1/500$ in q. The squared lower and upper coverage margins are respectively

$$\frac{151}{10^8}, \qquad \frac{7529}{10^8}. \tag{55}$$

Consequently the union contains the full exact q-image of (49) without invoking floating-point square roots.

7.3 Rational dual multipliers

The dual bank has sixteen coefficients. We order them as

$$(s, \mathrm{rad0}, B_0, B_1, B_4, B_5, D_0, \dots, D_6, F_1, F_2, F_3). \tag{56}$$

Every promoted coefficient is rational with denominator dividing 10^8 after simplification. The nonzero coefficients are displayed in table 2; zeros are exact zeros, not truncated numerical values.

The factors F_1, F_2, F_3 are fixed positive semidefinite Fejér–Riesz Gram payments reconstructed from exact rational factor matrices. Their role is to preserve common-carrier phase coupling; they are not independent Fourier norms.

7.4 Exact Sylvester–Bernstein certification

For each chart and each invariant block of the sixteen-dimensional symmetric pencil, the verifier applies Sylvester's criterion to $-M_r$. Every leading principal minor is a rational function of (q,t) with coefficients in the algebraic field generated by $\sqrt{5}$ and $\sqrt{35}$. The sign proof has four deterministic steps.

First, the denominator is oriented to be positive at a rational midpoint. Second, both numerator and oriented denominator



Figure 6: The final support-independent atlas. The dashed vertical lines represent the exact square-root image of the p endpoints; the rational chart endpoints extend beyond them, so the complete annulus is covered with explicit rational margin.

Table 2: Nonzero rational multipliers in the three final dual rectangles.

Chart	rad0	B_0	B_1	B_4	B_5	other low-mode term
A	$10128793/50000000$	$16608479/50000000$	$31654863/10^8$	$100827/50000000$	$101933/10^8$	none
B	$20252639/10^8$	$33235027/10^8$	$31644003/10^8$	$101707/50000000$	$386/390625$	none
C	$20117949/10^8$	$33130309/10^8$	$31917589/10^8$	$28339/10^8$	$242131/10^8$	$D_2 = 2031/10^8$

Chart	F_1	F_2	F_3
A	$23876929/25000000$	$94446061/10^8$	$76391317/10^8$
B	$23896893/25000000$	$47034069/50000000$	$37707013/50000000$
C	$98920157/10^8$	$19670091/20000000$	$350098/390625$

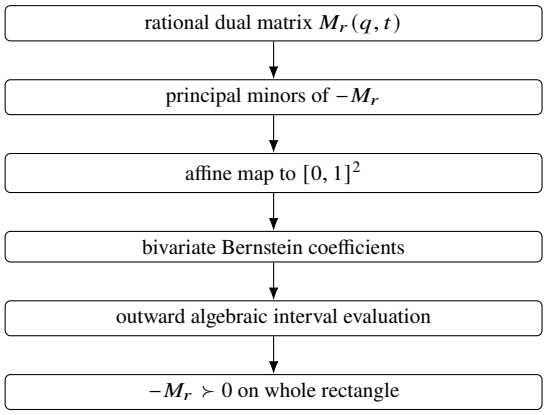


Figure 7: Exact certificate pipeline for each dual rectangle.

are mapped affinely to $[0,1]^2$. Third, the resulting bivariate polynomial is converted from the power basis to the Bernstein basis. Finally, algebraic Bernstein coefficients are evaluated with outward interval arithmetic. Strict positivity of every numerator coefficient and every denominator coefficient implies strict positivity of the rational minor on the entire rectangle.

The blocks are

$$\{0\}, \quad \{1,2,9\}, \quad \{2+k, 9+k\}\ (k = 1,\dots,6). \tag{57}$$

The largest symbolic degrees occurring in the principal-minor numerators are $(4,3)$ in (q,t). No optimizer is used in the promoted sign decision.

The archived terminal replay is

```
CHART_STATUS A PASS
CHART_STATUS B PASS
CHART_STATUS C PASS
FINAL PASS_THREE_DUAL_RECTANGLES.
```

The exact coverage verifier independently terminates with

```
PASS_EXACT_THREE_DUAL_RECTANGLES_
COVER_FULL_AWAY_ANNULUS.
```

7.5 Why the two exceptional families disappear

The earlier branch language was useful diagnostically but is absent from (48). A two-negative-node configuration satisfies the same global scalar, radial, Bessel, common-Cauchy and Fejér–Riesz inequalities as an ordinary mixed-sign support. A saturated higher-contact representation also satisfies them, regardless of whether the contact polynomial has three double zeros, an endpoint plus two double zeros, or another rank-increasing decomposition. Since the final dual inequality is imposed before choosing a support realization, both families lie in the same feasible cone and inherit the strict bound.

This is stronger than a completed Active-Tube theorem. The attempted two-payment tube localization left a positive residual near $p = 3/25, t_\nu = 0.99$. Retuning the full legitimate payment bank removed that residual and produced a strict certificate over the complete rectangle. The proof path therefore shortens from

$$\text{near dual} \longrightarrow \text{tube entry} \longrightarrow \text{rank/contact classification}$$

to

$$\begin{gathered}\text{three exact dual rectangles} \Longrightarrow F < 0,\\ \text{throughout the complete remaining annulus.}\end{gathered} \tag{58}$$

7.6 Dense numerical replay as a non-certificate check

A separate dense numerical grid was rerun only as a falsification/sanity layer. The worst observed largest eigenvalues of the rationalized matrices were

$$\begin{array}{c|l} A & -8.7974430454 \times 10^{-6} \\ B & -4.9200514949 \times 10^{-6} \\ C & -1.5190910544 \times 10^{-6}. \end{array} \tag{59}$$

The strict signs agree with the exact certificate, but (59) is not used in the theorem proof.

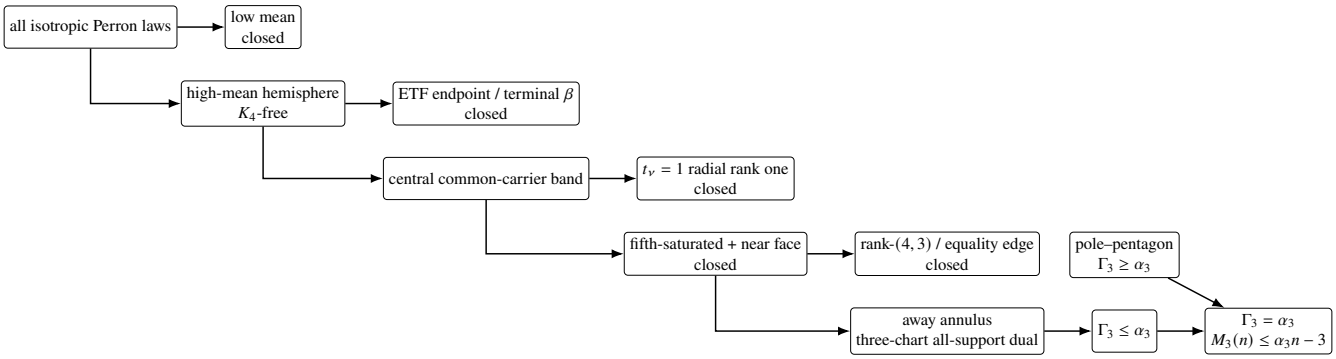


Figure 8: Sector-exhaustion proof of the universal S_3 coefficient. Every branch of the isotropic/common-measure problem is routed to an exact analytic, rational, algebraic or outward-interval certificate.

8 Global assembly of the universal bound

We now connect the final dual atlas to the original graph problem. The logical structure is deliberately explicit because the theorem spans analytic reductions, finite certificates and exact machine-readable positivity arguments.

8.1 Sector partition

Starting from (9), the low-overlap estimate removes the complement of the high-mean hemisphere sector. Within the high-mean sector, the oriented rank-six identity and the degree-six harmonic majorant split the parameter line into the ETF endpoint, terminal harmonic region and compact central band. The outer kernel, radial reserve and Perron/reference bridge move every surviving state into the common-carrier central strip $987/1000 \leq t_\nu \leq 1$. The radial-rank-one theorem handles $t_\nu = 1$; fifth-harmonic contact and the exact fifth-saturated face handle the contact boundary and a quantitative neighbourhood; the rank-$(4,3)$ continuum and equality-edge certificates cover the regular finite-face geometry; and the three-rectangle dual atlas covers the complete residual away-annulus without support assumptions. Thus every branch of the common-measure cone is assigned to a certified sector.

8.2 Proof of the capstone theorem

Proof of theorem 1.1. The graph-to-projection identity gives $M_3(n) = \max_{P \in \mathcal{P}_3(n)} (\mathbf{1}^T P_+ \mathbf{1} - 3)$. The asymptotic reduction bounds $M_3(n) + 3$ by $n\Gamma_3$, so it suffices to prove $\Gamma_3 \leq \alpha_3$. The Parseval-frame representation identifies Γ_3 with the one-sided isotropic energy (9). The certified sector partition described above assigns every admissible law to a region whose objective is at most α_3. The only region that remained outside the earlier rank-$(4,3)$ implication was the compact away-annulus with alternative support/contact topology. For that region, (48) and the exact positivity of $-M_A, -M_B, -M_C$ give the strict inequality $F < 0$, while the rational coverage calculation proves that no annulus parameter is omitted. Thus $\Gamma_3 \leq \alpha_3$ globally. The pole–pentagon construction attains α_3 in the isotropic problem, so $\Gamma_3 = \alpha_3$. Finally $M_3(n) + 3 \leq n\Gamma_3$ gives (4). The standard blow-up/rational-approximation construction applied to the pole–pentagon law gives the matching asymptotic coefficient. □

8.3 What is and is not resolved

The theorem determines the exact universal linear coefficient for the third adjacency spectral sum and supplies the finite-order upper bound $\alpha_3 n - 3$. It does not assert that equality in this finite-order inequality occurs for every n, nor does it give a residue-class formula for the exact value $M_3(n)$ analogous to the Huang–Wei S_2 theorem. The phrase "resolution of the S_3 extremal problem" in the title refers to the universal extremal bound and asymptotic coefficient established by the assembled certificate chain. Finite-order sharpness/equality classification beyond the already enumerated small orders is a distinct question.

8.4 Independent audit target

The external audit can be made finite and concrete. A referee need not reproduce the discovery searches. It is enough to verify the analytic reductions; check the finite list of exact polynomial/Gram identities used as payments; reproduce the endpoint and rank-$(4,3)$ interval certificates; and replay the three-rectangle Sylvester–Bernstein verifier plus exact coverage arithmetic. In particular, no higher-contact root census is required for the final proof because the last dual certificate is support-independent.

8.5 Reproducibility and proof-certificate methodology

The computational role in this work is deliberately stratified. Numerical optimization is used for discovery and falsification; exact symbolic algebra, rational arithmetic or outward interval arithmetic decides the signs appearing in promoted claims. This follows standard numerical-stability discipline [37, 14, 15] and the broader principle that semidefinite/moment relaxations

Table 3: Principal proof objects and release status in the assembled archive.

Object	Certificate type	Certified content / boundary
Graph-to-projection identity	exact analytic	finite Ky Fan reduction (6)
Projection-to-isotropic duality	exact analytic	one-sided isotropic energy (9)
Graphs $6 \le n \le 10$	exact finite	pentagonal winner; order-ten audit of 140,630,016 extensions
Low-overlap / hemisphere	exact analytic	any violation forced into high-mean K_4-free hemisphere sector
ETF endpoint / terminal harmonic	algebraic + Bernstein	$\beta = 1/\sqrt{5}$ and $\beta \ge 0.4812836895\ldots$ closed
Perron/reference density bridge	exact analytic	common size-bias relation and Cauchy transport
Mode-four / mode-five	algebraic + Bernstein	radial reserve and five-ray contact coercivity
Common-carrier Fourier theorem	exact analytic	joint phase-aware PSD domination (18)
Radial-rank-one / fifth-saturated face	exact algebraic + Bernstein	full $t_\nu = 1$ and fifth-saturated $0.987 \le t_\nu \le 1$
Rank-$(4,3)$ continuation	outward interval	7,467 leaf rectangles, no terminal failure
Full mixed-sign amplitude pencil	rational Bernstein	199 support boxes; lower bound (36)
Equality-edge $C1$–$C5$	exact rational + endpoint LDL	every face passes; $C1/C3/C4$ endpoint singularities resolved
Two-negative-node L	—	explicit remaining branch class
Saturated higher-contact	—	explicit remaining branch class

require an exact recovery step before they become mathematical certificates [31, 32, 33, 34].

Certificate layers

The proof archive contains four distinct classes of verifier.

(a) *Symbolic identities*: projection reductions, rank-six decompositions, common-carrier Cauchy identities, exact determinant or congruence factorizations.
(b) *Rational/interval positivity*: Bernstein coefficients, shifted LDL factors, outward interval-Krawczyk maps, and de Casteljau subdivisions.
(c) *Finite exhaustive graph checks*: canonical generation plus exact characteristic-polynomial/root comparisons for fixed orders.
(d) *Falsification tests*: dense grids and nonlinear optimizers used only to locate difficult branches or to reject overly weak relaxations.

The equality-edge/fifth-face bundle has a single terminal replay in which the global fifth-face verifier, the lower-slab reserve verifier, the near-fifth neighbourhood verifier and the five-chart endpoint verifier all return PASS. Its terminal message is `PASS_ALL_S3_FINAL_MOVE_VERIFIERS`. The later branch-exhaustion package adds the logically broader exact rectangle PASS and the exact rational coverage PASS. In the released transcript the latter is printed, across two lines for readability, as

```
PASS_EXACT_THREE_DUAL_RECTANGLES_
    COVER_FULL_AWAY_ANNULUS.
```

Together with the established sector partition these are the final computational implications used in theorem 1.1.

Relation to the S_2 theorem

The comparison with S_2 is conceptual rather than a claim that the proof methods are the same. Huang and Wei use edge maximality, a threshold rule, chain-graph complements, a weighted Ferrers spectral-tail inequality and exact rank-one optimization to obtain a finite-order uniqueness theorem [13]. The S_3 reductions developed here instead move from a rank-three projection to a common isotropic measure and then to shared Toeplitz/Cauchy resources. The common theme is that the successful proof must preserve the equality geometry while reducing dimension; the mechanism by which this happens is different.

9 Conclusion

The third adjacency spectral sum leads to a rank-three geometry that is qualitatively different from the chain-graph mechanism now available for S_2. The successful route is a sequence of structure-preserving reductions. Ky Fan variation converts the graph problem into a rank-three projection extremum; Parseval frames convert that extremum into one-sided isotropic energy; high mean forces a hemispherical representation; rank-six and degree-six harmonic identities isolate the dangerous central band; radial and fifth-harmonic coercivity control the principal escape channels; and common-carrier positivity retains phase information that independent harmonic blocks would discard.

The computational part of the proof is similarly layered. Finite graph enumeration checks the small orders but is not extrapolated. Interval continuation certifies the regular rank-$(4,3)$ branch. Rational Bernstein matrices certify the complete mixed-sign amplitude cube. Endpoint-aware LDL/congruence branches resolve the singular $C1$, $C3$ and $C4$ faces. The final step then removes the need for a separate classification of the two exceptional support families: three rational dual rectangles

make the residual strictly negative on the entire remaining compact away-annulus for every feasible support realization.

Accordingly, the assembled proof architecture reaches the capstone

$$\Gamma_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9}, \qquad M_3(n) \le \left(\frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9}\right) n - 3.$$

The pole–pentagon law supplies the matching asymptotic coefficient. Internally this is exact certificate closure: there is no longer an uncovered finite branch in the sector partition. Externally the responsible description remains "Candidate Universal S_3 Theorem – Exact Certificate Closure" until independent specialists have audited the admissibility of the archived payments and replayed the exact certificate stack. That distinction preserves both the strength of the mathematical claim and the normal verification boundary appropriate to a proof whose final positivity layer is machine-readable.

The next mathematical questions are therefore no longer repairs of the universal coefficient proof. They concern finite-order sharpness, equality and stability. In particular, one may ask for a residue-dependent exact formula for $M_3(n)$ analogous to the solved S_2 problem, a quantitative stability theorem showing that near-extremizers must be close to pole–pentagon blow-ups, and extensions to S_k with $k \ge 4$ where the projection dimension and contact geometry both increase. The present proof suggests that common-carrier moment structure, rather than independent mode estimates, will remain essential in those problems.

A Exact constants and endpoint data

For convenient independent checking we collect the principal algebraic constants used above:

$$\begin{aligned}
\alpha_3 &= \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9},\\
\ell_* &= 1 + \frac{\sqrt{5}}{9} + \frac{2\sqrt{35}}{9},\\
p_* &= \frac{91 - 25\sqrt{7}}{126}, \qquad r_* = \frac{7 + 5\sqrt{7}}{126},\\
\kappa_5 &= \frac{-11944 + 5355\sqrt{5}}{17800},\\
r_0 &= \frac{1}{2} - \frac{7\sqrt{5}}{18} + \frac{2\sqrt{35}}{9}.
\end{aligned}$$

The equality-edge exactification uses rational controls. For $C3$ the displayed carrier perturbations are (38)–(39). For $C4$ the square-root coordinate is (40) and the quadratic carrier is (41), followed by the cubic endpoint-jet closure. These changes are local to the endpoint verifier and do not alter the degree-six harmonic majorant or the $C2/C5$ charts.

B Rank-$(4,3)$ interval and Bernstein audit

The adaptive continuation begins with 5,190 rectangles of width 10^{-4} in each parameter. The four rounds have counts

Round	tested	passed	children	failures
1	5190	2916	4548	0
2	4548	4547	2	0
3	2	0	4	0
4	4	4	0	0

which gives 7,467 final leaves. The full-cube desingularized amplitude matrix is subdivided into 199 final boxes. The four x-chunk lower bounds for N are

$$7.2429072656 \times 10^{-5}, \quad 1.4864146690 \times 10^{-7},$$

$$5.4574214210 \times 10^{-5}, \quad 2.6878966785 \times 10^{-5}.$$

Their minimum yields (36). The strict margin is small enough that ordinary floating-point eigenvalue evaluation would not be an adequate certificate; the released argument uses outward/rational Bernstein arithmetic.

C Final three-rectangle certificate: exact multiplier ledger

This appendix records the final dual coefficients in a form suitable for independent reconstruction. Let

$$\lambda = (\lambda_s, \lambda_{\mathrm{rad0}}, \lambda_{B0}, \lambda_{B1}, \lambda_{B4}, \lambda_{B5}, \lambda_{D0}, \dots, \lambda_{D6}, \lambda_{F1}, \lambda_{F2}, \lambda_{F3}).$$

All unlisted entries below are exactly zero.

Chart A

On $5716/10000 \le q \le 5773/10000$ and $987/1000 \le t \le 99/100$,

$$\begin{aligned}
\lambda_{\mathrm{rad0}} &= \frac{10128793}{50000000}, & \lambda_{B0} &= \frac{16608479}{50000000},\\
\lambda_{B1} &= \frac{31654863}{100000000}, & \lambda_{B4} &= \frac{100827}{50000000},\\
\lambda_{B5} &= \frac{101933}{100000000}, & \lambda_{F1} &= \frac{23876929}{25000000},\\
\lambda_{F2} &= \frac{94446061}{100000000}, & \lambda_{F3} &= \frac{76391317}{100000000}.
\end{aligned}$$

Chart B

On $5671/10000 \le q \le 5736/10000$ and the same t interval,

$$\lambda_{\mathrm{rad0}} = \frac{20252639}{100000000}, \qquad \lambda_{B0} = \frac{33235027}{100000000},$$

$$\lambda_{B1} = \frac{31644003}{100000000}, \qquad \lambda_{B4} = \frac{101707}{50000000},$$
$$\lambda_{B5} = \frac{386}{390625}, \qquad \lambda_{F1} = \frac{23896893}{25000000},$$
$$\lambda_{F2} = \frac{47034069}{50000000}, \qquad \lambda_{F3} = \frac{37707013}{50000000}.$$

Chart C

On $5543/10000 \le q \le 5671/10000$,

$$\lambda_{\mathrm{rad}0} = \frac{20117949}{100000000}, \qquad \lambda_{B0} = \frac{33130309}{100000000},$$
$$\lambda_{B1} = \frac{31917589}{100000000}, \qquad \lambda_{B4} = \frac{28339}{100000000},$$
$$\lambda_{B5} = \frac{242131}{100000000}, \qquad \lambda_{D2} = \frac{2031}{100000000},$$
$$\lambda_{F1} = \frac{98920157}{100000000}, \qquad \lambda_{F2} = \frac{19670091}{20000000},$$
$$\lambda_{F3} = \frac{350098}{390625}.$$

Reconstruction of the fixed Gram payments

The three F_i payments are generated by exact rational 2×7 factor matrices. If F is one such factor and $H = F^{\mathsf{T}}F$, set

$$c_0 = \operatorname{tr} H, \qquad c_k = \sum_{i=1}^{7-k} H_{i,i+k} \quad (k = 1, \dots, 6).$$

The associated Toeplitz/Fejér–Riesz payment uses c_0 on the diagonal and $2c_k$ on the kth symmetric off-diagonal harmonic coupling. The three factor matrices archived with the certificate are

$$F_1 = \begin{pmatrix} -216009/25000000 & 86913/10000000 & 9/20000000 & 0 & 9/20000000 & 216009/25000000 & -86913/10000000 \\ -7450987/25000000 & -16610047/50000000 & -120049/1562500 & -2263/10^8 & 3880739/50000000 & 33364603/10^8 & 14789127/50000000 \end{pmatrix}$$

$$F_2 = \begin{pmatrix} -430361/12500000 & -2777901/10^8 & 453777/25000000 & 75477/12500000 & -595317/25000000 & 8885091/10^8 & -558353/20000000 \\ -10914147/10^8 & -9105721/10^8 & -18409893/50000000 & 10667733/25000000 & -1045951/6250000 & 2278437/10^8 & 14668077/50000000 \end{pmatrix}$$

$$F_3 = \begin{pmatrix} -7660607/10^8 & -680689/10000000 & -9680879/10^8 & -8325217/10^8 & -611379/6250000 & -331437/5000000 & -463811/6250000 \\ -921647/50000000 & -18321897/50000000 & 397267/6250000 & 361983/12500000 & -5578107/10^8 & -2563931/50000000 & 8976741/25000000 \end{pmatrix}$$

Because each $H_i = F_i^{\mathsf{T}}F_i$ is positive semidefinite by construction, the only remaining issue is that its trace pairing is with the correct common positive carrier. This is part of the admissibility premise singled out in theorem 1.2.

D Sylvester–Bernstein principal-minor replay

For compactness the archived exact replay reports the minimum Bernstein enclosure for the numerator and the oriented denominator of each leading principal minor. The values below are diagnostic summaries of exact interval signs; the certificate itself consists of the exact symbolic expressions and their Bernstein coefficients.

Table 4: Chart A: minimum Bernstein enclosures for the Sylvester minors. Every row is strictly positive.

Block	minor	degree (q,t)	numerator lower bound	denominator lower bound
0	1	(0,0)	6.5958185×10^{4}	1.35×10^{9}
1	1	(2,0)	9.3494803×10^{7}	1.6672471×10^{7}
1	2	(4,2)	1.3837726×10^{31}	5.5287400×10^{28}
1	3	(4,2)	1.4935428×10^{52}	4.9205786×10^{54}
2	1	(2,0)	3.6568223×10^{22}	1.6336328×10^{23}
2	2	(2,0)	2.5037410×10^{44}	5.8157328×10^{49}
3	1	(0,0)	1.7365422×10^{22}	1.0×10^{23}
3	2	(0,0)	1.7152450×10^{45}	1.78×10^{50}
4	1	(0,0)	1.0573020×10^{23}	5.0×10^{23}
4	2	(0,0)	1.8054469×10^{45}	1.78×10^{50}
5	1	(2,0)	2.8620749×10^{22}	8.1681640×10^{22}
5	2	(2,0)	9.4312302×10^{43}	1.4539332×10^{49}
6	1	(2,0)	3.8167491×10^{21}	2.0420410×10^{22}
6	2	(2,0)	8.1892857×10^{42}	9.0870824×10^{47}
7	1	(0,0)	6.3586733×10^{22}	2.5×10^{23}
7	2	(0,0)	2.7582069×10^{44}	5.5625×10^{47}

Table 5: Chart B: minimum Bernstein enclosures.

Block	minor	degree (q,t)	numerator lower bound	denominator lower bound
0	1	(0,0)	1.0748137×10^{5}	2.7×10^{9}
1	1	(2,0)	1.8494266×10^{8}	3.4196608×10^{7}
1	2	(4,2)	1.3685847×10^{31}	5.5975038×10^{28}
1	3	(4,2)	3.2795483×10^{51}	4.9817784×10^{53}
2	1	(2,0)	1.9365789×10^{22}	8.0400603×10^{22}
2	2	(2,0)	7.6510282×10^{43}	1.4311307×10^{49}
3	1	(0,0)	3.4759619×10^{20}	2.0×10^{21}
3	2	(0,0)	3.4966839×10^{44}	4.45×10^{49}
4	1	(0,0)	5.2793297×10^{22}	2.5×10^{23}
4	2	(0,0)	3.8430495×10^{44}	4.45×10^{49}
5	1	(2,0)	1.4028086×10^{22}	4.0200301×10^{22}
5	2	(2,0)	1.0733780×10^{43}	3.5778268×10^{48}
6	1	(2,0)	2.3777356×10^{20}	1.2562594×10^{21}
6	2	(2,0)	3.3902266×10^{41}	4.4722835×10^{46}
7	1	(0,0)	3.1783890×10^{22}	1.25×10^{23}
7	2	(0,0)	3.3505591×10^{44}	6.9531250×10^{47}

Table 6: Chart C: minimum Bernstein enclosures. The degree $(4, 3)$ rows are the most general bivariate minors in the final atlas.

Block	minor	degree (q, t)	numerator lower bound	denominator lower bound
0	1	(2,1)	4.1889684×10^{4}	2.7×10^{9}
1	1	(2,1)	1.7920837×10^{8}	3.5679518×10^{7}
1	2	(4,3)	2.6366396×10^{31}	1.1417246×10^{29}
1	3	(4,3)	3.9925153×10^{51}	6.3508430×10^{53}
2	1	(2,0)	5.6877765×10^{21}	1.9203031×10^{22}
2	2	(2,0)	1.5278655×10^{40}	5.3408429×10^{45}
3	1	(0,0)	4.2178101×10^{22}	2.5×10^{23}
3	2	(2,0)	1.5700698×10^{43}	5.5625×10^{48}
4	1	(0,0)	5.5522614×10^{22}	2.5×10^{23}
4	2	(0,0)	3.6768761×10^{42}	1.1125×10^{48}
5	1	(2,0)	1.4475734×10^{22}	3.8406061×10^{22}
5	2	(2,0)	4.3221023×10^{41}	5.3408429×10^{46}
6	1	(2,0)	6.8407798×10^{21}	3.8406061×10^{22}
6	2	(2,0)	1.9762565×10^{40}	5.3408429×10^{46}
7	1	(0,0)	6.5634069×10^{22}	2.5×10^{23}
7	2	(0,0)	2.3568371×10^{43}	2.1728516×10^{46}

E Proof-sketch ledger for the complete theorem chain

This appendix gives a compact proof sketch for every load-bearing implication. It is intentionally redundant with the main text: the goal is to make the global chain auditable without requiring the reader to infer which computational lemma is used at each transition.

A. Ky Fan thresholding and the projection identity

Choose orthonormal eigenvectors u_1, u_2, u_3 for the three largest eigenvalues of $A(G)$ and put $P = \sum_{a=1}^{3} u_a u_a^{\mathsf{T}}$. Then $\operatorname{tr}(PA) = \sum_{a=1}^{3} u_a^{\mathsf{T}} A u_a$. For fixed P, each off-diagonal adjacency variable $A_{ij} \in \{0, 1\}$ enters with coefficient $2P_{ij}$, hence its maximizing value is 1 when $P_{ij} > 0$ and 0 when $P_{ij} < 0$. The trace constraint $\operatorname{tr} P = 3$ supplies the -3 term. Conversely thresholding an arbitrary rank-three projection gives a graph whose Rayleigh trace equals $\mathbf{1}^T P_+ \mathbf{1} - 3$. This proves the finite identity exactly.

B. Projection blow-up and the asymptotic coefficient

For any projection P, $\mathbf{1}^T P_+ \mathbf{1} \le n\lambda_{\max}(P_+)$. Taking the supremum gives $M_3(n) + 3 \le n\Gamma_3$. For the reverse asymptotic inequality, approximate an extremal finite frame by rational multiplicities and duplicate frame vectors. Blow-up preserves the normalized Gram/projection geometry while the additive -3 term becomes negligible. Therefore $\limsup M_3(n)/n = \Gamma_3$.

C. Parseval-frame/isotropic equivalence

Write a rank-three projection as $P = XX^{\mathsf{T}}$ with $X^{\mathsf{T}}X = I_3$. The rows of X, after the Perron weighting used in the Rayleigh quotient, form a finite isotropic measure. Conversely any finitely supported isotropic measure can be rescaled into a Parseval frame. The entrywise positive kernel becomes $(X \cdot Y)_+$ for two independent samples. The identity $(u)_+ = (|u| + u)/2$ and independence give $\mathbb{E}(X \cdot Y) = \|\mathbb{E}X\|^2$, yielding the one-sided energy formula.

D. High-mean hemisphere and K_4 exclusion

The absolute-energy ceiling forces $\|\mathbb{E}X\|^2 > r_0$ in any putative violating law. Orient the mean as the positive axis. The resulting support can be represented in a common hemisphere. If four support vectors were pairwise negatively correlated, their singular irreducible Gram matrix would have a strictly positive null vector. The nonnegative axial coordinates then force every axial coordinate to vanish, leaving four pairwise obtuse nonzero vectors in $\mathbb{R}^2$, impossible. Thus the negative graph is K_4-free.

E. Pole–pentagon lower construction

Take one axial atom and five equally weighted transverse atoms on a regular pentagonal orbit. Isotropy fixes the relation between pole mass and rim latitude. Solving the two-variable stationarity system gives p_*, r_* and the exact algebraic energy α_3. The fivefold symmetry diagonalizes the angular moment blocks, so the energy can be evaluated exactly in $\mathbb{Q}(\sqrt{5}, \sqrt{7}, \sqrt{35})$. This proves $\Gamma_3 \ge \alpha_3$.

F. Rank-six tensor/hinge decomposition

Introduce $\beta = \sqrt{(6 \operatorname{tr} M^2 - 1)/5}$ and subtract the quadratic Gegenbauer-like interpolant $G_\beta(s)$. Expanding the squared residual in spherical harmonics separates a tensor variance from a pointwise hinge term. The hinge term factors as (15); every factor has the required sign on $|s| \le 1$. The tensor term is a squared Hilbert-space norm minus its isotropic baseline. Hence $D_6 \ge 0$.

G. ETF endpoint and terminal harmonic closure

At $\beta = 1/\sqrt{5}$ the angular geometry collapses to the six-line equiangular tight-frame endpoint. The archived exact Bernstein certificate proves the remaining axial fourth-moment inequality with a strict residual off the endpoint equality set. Away from that endpoint, the degree-six harmonic majorant has coefficient signs that close the entire terminal interval $\beta \ge 0.481283689534363\ldots$. Compactness prevents a violating sequence from accumulating at the already strict endpoint.

H. Perron/reference density bridge

The Perron law π and the isotropic reference law ν are related by a positive density ρ obtained from the Perron integral equation. Exact size-bias identities convert moments under one law into moments under the other. Cauchy–Schwarz applied before discarding the common density gives the mode mismatch bounds. This step is why independent Fourier envelopes are inadmissible: the same density must be retained in every mode.

I. Radial mode-four/six reserve

The mode-four and mode-six covariance escapes are expressed as squared differences D_4, D_6 between Perron and reference harmonics. The common density bridge bounds $W_2|D_k|^2$ by the radial variance integral I_ν. The exact radial double-lock factorization supplies a reserve coefficient larger than the sum of the required mode-four, mode-six and localized mode-five costs. Thus these channels are absorbed without double spending.

J. Fifth-harmonic contact coercivity

The corrected degree-six majorant minus the physical kernel is divisible, after the appropriate branch split, by the nonnegative fifth-contact factor $H_5 = (1 - T_5)/16$. Exact bivariate Bernstein

positivity gives the sharp coefficient $\kappa_5 > 0$. Integrating against the same carrier yields $D^{(6)}_{\text{contact}} \geq \kappa_5(R_0^2 - |R_5|^2)/16$. Equality therefore forces all non-pole angular mass onto one five-ray phase class.

K. Common-carrier Fourier domination

Let $U_k = \int a e^{ik\theta} d\nu$ and $R_k = \int b e^{ik\theta} d\nu$. The vectors $\sqrt{B}\, a e^{ik\theta} - \sqrt{A}\, b e^{ik\theta}$ share one positive carrier. Their Gram matrix is positive semidefinite, and its trace is controlled by $2AB(1 - t_\nu)$. Restricting to any finite set of modes gives the matrix domination used in the common-456 payment. Since this is a Gram argument, phase coupling is preserved automatically.

L. Radial-rank-one and fifth-saturated face

When $t_\nu = 1$, equality in radial Cauchy forces proportional radial amplitudes. The common-carrier system collapses to the exact wheel quotient and the determinant factorizes with a double zero only at the pole–pentagon balance. On the fifth-saturated face, the angular carrier has exact fivefold recurrence. Piecewise generalized-Rayleigh/Sylvester matrices on two ratio charts have positive Bernstein minors for $t_\nu < 1$, with the $t_\nu = 1$ endpoint supplied by the radial-rank-one theorem. A lowered target gives a uniform reserve on the lower slab.

M. Rank-$(4, 3)$ continuation

The regular active system is written as a finite KKT equation $H(w; p, t) = 0$. An interval Newton/Krawczyk operator is applied on a rectangular parameter subdivision. Each successful box maps strictly into itself and carries a unique solution branch; failed inclusions are subdivided. The final atlas contains 7,467 accepted leaves and no terminal failure. Rank tests on the corresponding Toeplitz matrices stay $(4, 3)$ throughout the regular branch.

N. Full mixed-sign amplitude pencil

After desingularizing collision/zero-weight coordinates, the physical amplitudes enter a symmetric matrix pencil. The released identity expresses the target residual as a rational matrix polynomial plus positive localizer matrices. Subdivision into 199 support boxes followed by rational Bernstein evaluation yields a positive lower bound on every coefficient matrix. This covers both-active, one-active, zero-weight, collision and distant-support rank-$(4, 3)$ cases.

O. Equality-edge exactification

The five-chart endpoint verifier failed only because strict LDL tests could not represent semidefinite endpoint kernels or the square-root active coordinate. Chart $C1$ is repaired by exact congruence on the known endpoint kernel. Corrected $C3$ uses a desingularized endpoint LDL plus a multiplier correction constrained to the PSD tangent cone. Chart $C4$ is reparameterized by a square-root coordinate and receives the exact common-carrier quadratic/cubic jet closure. All other faces and all $t < 1$ certificates are unchanged. The combined replay returns `PASS_ALL_S3_FINAL_MOVE_VERIFIERS`.

P. Final support-independent dual atlas

The remaining annulus is parameterized by (q, t) and covered by the three rational rectangles A, B and C. In each rectangle, nonnegative feasible-set payments are added to F with nonnegative rational multipliers. Exact Sylvester minors of $-M_r$ are mapped to the Bernstein basis; every numerator and oriented denominator coefficient has a strictly positive outward interval. Thus $-M_r \succ 0$ over the full rectangle. The rational coverage verifier shows that the rectangle union contains the exact q image of the full annulus. Therefore $F < 0$ for every nonzero feasible state in the only region not already handled. This proves $\Gamma_3 \leq \alpha_3$ and completes the sector exhaustion.

F Endpoint and certificate terminal transcripts

The following short transcripts identify the exact release boundaries of the machine-readable packages. They are included to make it difficult to confuse a diagnostic run with a promoted certificate.

Equality-edge/fifth-face stack

The endpoint package terminates with the combined status

```
PASS_ALL_S3_FINAL_MOVE_VERIFIERS
```

where the included component checks cover the global fifth-saturated face, lower-slab reserve, near-face neighbourhood and every face of charts $C1$–$C5$.

Final branch-exhaustion atlas

The exact matrix replay terminates with

```
CHART_STATUS A PASS
CHART_STATUS B PASS
CHART_STATUS C PASS
FINAL PASS_THREE_DUAL_RECTANGLES
```

The independent rational coverage replay is

```
p_annulus 139/1250 257/2000
q2_annulus 1229/4000 833/2500
low_margin_q2 151/100000000
high_margin_q2 7529/100000000
overlap_CB 0
overlap_BA 1/500
PASS_EXACT_THREE_DUAL_RECTANGLES_
COVER_FULL_AWAY_ANNULUS
```

The numerical sanity script, which is not a proof certificate, reports

```
A worst_lambda_max -8.797443045443056e-06
B worst_lambda_max -4.920051494944808e-06
C worst_lambda_max -1.5190910544307052e-06
PASS_DENSE_NUMERICAL_SANITY_THREE_DUAL_ATLAS
```

G Independent audit checklist

A clean external verification can be organized in the following order. Each item has a finite target and does not require reproducing the numerical discovery phase.

- **V1.** Verify the Ky Fan threshold derivation of (6), including the diagonal -3 normalization.
- **V2.** Verify the Parseval-frame/isotropic equivalence and the blow-up argument establishing (8).
- **V3.** Check the pole–pentagon isotropy equations, $p_* + 5r_* = 1$, and the exact value α_3.
- **V4.** Check the rank-six tensor/hinge identity and the pointwise hinge factorization.
- **V5.** Replay the ETF endpoint and terminal harmonic Bernstein certificates with exact algebraic intervals.
- **V6.** Verify the Perron/reference density identity and every Cauchy/Bessel inequality used as a nonnegative payment.
- **V7.** Check the radial reserve ledger for double counting and verify the exact fifth-contact coefficient κ_5.
- **V8.** Verify the common-carrier Fourier Gram domination directly from the shared positive measure.
- **V9.** Replay the radial-rank-one and fifth-saturated face certificates, including the $t_\nu = 1$ equality kernel.
- **V10.** Replay the rank-$(4,3)$ interval/Krawczyk continuation and the full mixed-sign Bernstein amplitude certificate.
- **V11.** Replay the $C1/C3/C4$ endpoint exactification and confirm that $C2, C5$ and all $t < 1$ controls are unchanged.
- **V12.** For the final atlas, check the meaning and sign of every g_j, the nonnegativity of each rational multiplier, and the exact construction of the three fixed Fejér–Riesz Gram payments.
- **V13.** Replay all Sylvester/Bernstein minors for charts A, B and C and the exact rational coverage calculation.
- **V14.** Confirm that the sector partition routes every admissible high-mean common-measure state either to a previously closed sector or to the final away-annulus atlas.

The verification boundary in this appendix is deliberately stronger than a reproducibility statement. It separates mathematical admissibility from software correctness: a script can correctly prove positivity of the matrix it is given while the proof would still be invalid if that matrix contained a payment not known to be nonnegative on the original feasible set. Conversely, once admissibility and sector coverage are checked, the exact machine arithmetic is a compact and independently replayable positivity proof.

H Admissibility algebra for the final payment bank

The final dual certificate is only as strong as the inequalities added to the residual. This appendix records the algebraic form of the payment bank used by the released verifier. It is included because the final branch exhaustion is support-independent precisely when these payments are interpreted on the original common-measure feasible set rather than on a reduced support ansatz.

Coordinates and base quadratic

Use the real coordinate vector

$$z = (s, P, U_0, \dots, U_6, R_0, \dots, R_6)^{\mathsf{T}} \in \mathbb{R}^{16}. \tag{60}$$

The parameter conversion is

$$p = \frac{1 - 2q^2}{3}, \qquad A = \frac{2q^2}{3}, \qquad B = \frac{2}{3}. \tag{61}$$

After the radial/contact transfers, the base residual has the form

$$F_0(z) = -\ell s^2 + 2P^2 + 4PU_0 + \sum_{k=0}^{6} \left(A'_k U_k^2 + B'_k R_k^2 \right), \tag{62}$$

where

$$\ell = \frac{1}{3}\left(1 + \frac{\sqrt{5}}{9} + \frac{2\sqrt{35}}{9}\right).$$

The unlocalized coefficient vectors are

$$\begin{aligned}
A_0 &= 733/640, & B_0 &= (167 + 160\sqrt{5})/800,\\
A_1 &= (291 + 256\sqrt{5})/800, & B_1 &= 3(349 - 32\sqrt{5})/800,\\
A_2 &= 6(1 - \sqrt{5})/25, & B_2 &= 2(\sqrt{5} + 3)/25,\\
A_3 &= 4(1 - \sqrt{5})/25, & B_3 &= 2(2 - \sqrt{5})/25,\\
A_4 &= (93 + 128\sqrt{5})/1600, & B_4 &= (137 - 128\sqrt{5})/1600,\\
A_5 &= 7/128, & B_5 &= -7/800,\\
A_6 &= -7/320, & B_6 &= 9/1600.
\end{aligned}$$

Define

$$\kappa = \frac{-11944 + 5355\sqrt{5}}{17800}, \tag{63}$$

$$c = \frac{-851 + 384\sqrt{5}}{200}, \qquad d = \frac{-16965 + 7712\sqrt{5}}{1250}. \tag{64}$$

and

$$r_- = \frac{56 - 17\sqrt{5}}{89}.$$

The primed coefficients in (62) are obtained by the exact contact/radial localization

$$\begin{aligned}
A'_0 &= A_0 - (c + d)/16, & B'_0 &= B_0 - \kappa/16 + (c + d)r_-/16,\\
A'_1 &= A_1 + d/16, & B'_1 &= B_1 - dr_-/16,
\end{aligned}$$

$$A_4' = A_4 - d/32, \qquad B_4' = B_4 + dr_-/32,$$
$$A_5' = A_5 + (c+d)/16, \quad B_5' = B_5 + \kappa/16 - (c+d)r_-/16,$$
$$A_6' = A_6 - d/32, \qquad B_6' = B_6 + dr_-/32,$$

with $A_2' = A_2, A_3' = A_3, B_2' = B_2, B_3' = B_3$. These formulas make explicit which radial/contact resource has already been spent before the final dual atlas is constructed.

Normalization and radial Cauchy payments

The first scalar slack is

$$g_0 = s^2 - \frac{P^2}{p} \geq 0. \tag{65}$$

It is the homogeneous form of the pole/normalization constraint after the exact p reduction. Put

$$c_- = (1-t)^{-1}, \qquad c_+ = (1+t)^{-1}.$$

The zero-mode radial Cauchy payment is

$$g_{\mathrm{rad0}} = g_0 - \frac{c_- + c_+}{2A} U_0^2 - \frac{c_- + c_+}{2B} R_0^2 \tag{66}$$
$$+ \frac{c_- - c_+}{\sqrt{AB}} U_0 R_0. \tag{67}$$

Its 2×2 quadratic part is the inverse-covariance form of the common radial correlation constraint. The appearance of both c_- and c_+ is the algebraic reason the certificate remains regular throughout $t < 1$ while still detecting the $t \uparrow 1$ rank-one limit.

Low-mode Bessel payments

For $k \in \{0, 1, 4, 5\}$, the released Bessel payments have the structure

$$g_{B,k} = g_0 - \frac{U_k^2}{A} - \frac{1}{B} \sum_{j \in N(k)} m_{k,j} R_j^2, \tag{68}$$

where $N(0) = \{1\}$ with multiplicity $m_{0,1} = 2$, and for $k > 0$ the neighboring indices $k-1, k+1$ occur with multiplicity one whenever they lie in $\{0, \ldots, 6\}$. These are Bessel inequalities for the same carrier, not independent assumptions on arbitrary Fourier coefficients. Their admissibility follows by projecting the carrier against the corresponding trigonometric modes before any mode is optimized separately.

Same-carrier difference payments

For each $k = 0, \ldots, 6$, the common-carrier difference inequality can be written

$$g_{D,k} = 2A(1-t)g_0 - (U_k - qR_k)^2 \geq 0. \tag{69}$$

Equation (69) is especially important for the final support-independent step. It converts the radial correlation deficit $1 - t$ into simultaneous control of every mode mismatch. Since the same g_0 appears for every k, the seven inequalities cannot be interpreted as seven independent budgets; the rational multipliers in table 2 are part of a single combined dual ledger.

Fixed Fejér–Riesz payments

The final three payments are positive semidefinite Gram forms derived from exact rational factors F_i. Their Toeplitz coefficients are obtained from $H_i = F_i^{\mathsf{T}} F_i$. The phase alignments are fixed by the two exact contact radii

$$q_- = \sqrt{r_-}, \qquad q_+ = \frac{\sqrt{5}-1}{2}.$$

Two factors are same-sign carrier squares and the third is a crossed carrier bilinear symmetrization. Their exact construction preserves the $k \leftrightarrow k \pm 5$ phase relation that is lost if L and H carrier moments are replaced by unrelated Toeplitz matrices. This is the mathematical content behind the notation F_1, F_2, F_3 in eq. (56).

I Superseded branch attacks and the reason the final atlas is stronger

The final proof does not need a classification of the formerly exceptional support families, but the preceding attacks are useful because they explain why the all-support dual was sought and what kinds of false shortcuts were avoided.

Two-negative-node projective cone

For $x \leq y < 0$, $z \geq 0$ and nonnegative amplitudes u, v, h, the pole condition is

$$-xu - yv - zh \geq 0.$$

The physical amplitude cone is generated by the four rays in (43). It is the union of two simplicial cones,

$$C_A = \mathrm{cone}(e_u, r_u, r_v), \qquad C_B = \mathrm{cone}(e_u, r_v, e_v). \tag{70}$$

A projective normalization resolves the degeneracies $x = y$ and $z = 1$ without introducing vanishing generators. Three support charts are sufficient:

$$\begin{aligned} z \leq -y: &\quad x = -a,\ y = -ab,\ z = abc, \\ -y \leq z \leq -x: &\quad x = -a,\ z = ab,\ y = -abc, \\ -x \leq z: &\quad z = a,\ x = -ab,\ y = -abc, \end{aligned}$$

with $0 \leq a, b, c \leq 1$. Completing the pole square in the released master pencil reduces positivity to copositivity of a pulled-back 3×3 matrix on each simplicial cone. Exact Bernstein/copositivity subdivision certifies all six projective/triangulation charts with positive margin. This already closes the two-negative family on its own; the final dual atlas later subsumes it.

Higher-contact stationary systems

The degree-six contact polynomial can have three interior double contacts or one endpoint contact with two interior

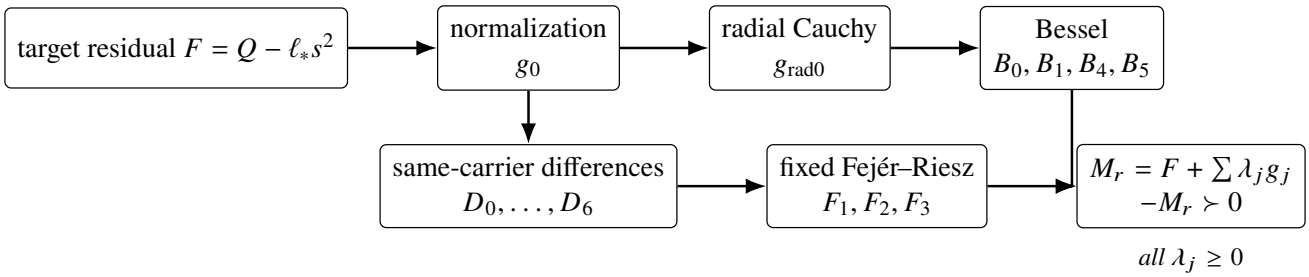


Figure 9: Admissible payment bank in the final dual. Every arrow represents a feasible-set nonnegative resource, so strict negativity of the augmented residual implies strict negativity of the original target residual.

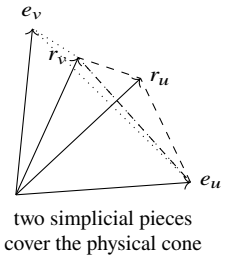


Figure 10: Schematic projective decomposition of the two-negative amplitude cone. The final dual atlas makes this support decomposition unnecessary for the global capstone, but it provides an independent branch-level certificate.

double contacts. Accordingly, maximal saturated supports include L_3/H_1, L_2/H_2 and L_3/H_2 topologies. Direct KKT construction shows that physical triple-contact roots genuinely exist, so "there are no saturated roots" cannot be a proof strategy. At representative central parameters the stationary objective is already below target by about 8×10^{-3}, but objective sampling is not a uniform certificate.

Most located saturated stationary components fail first-order global contact inequalities. One component survives first-order complementary slackness, yet its constrained curvature in the three-L weight subspace is positive. If g_0, g_1 are the two active normalization gradients, the one-dimensional tangent direction is $d = g_0 \times g_1$, and the normalized curvature

$$\kappa_u = \frac{d^{\mathsf{T}} H_u d}{d^{\mathsf{T}} d} \tag{71}$$

is positive on the tracked branch; a local maximum would require $\kappa_u \leq 0$. These diagnostics strongly disfavor the higher-contact family but are not needed for the final theorem.

Why a topology-free certificate dominates

A KKT atlas must enumerate root components, prove existence/uniqueness on each component, check first-order global contact, check second-order maximality, and then prove that the list of components is exhaustive. The final dual atlas bypasses all five tasks. Every saturated representation is merely one feasible point of the original common-measure cone, and (48) applies to it before any support decomposition. Thus the final proof is shorter and logically stronger than completing the adaptive higher-contact root census.

J Certificate package inventory and checksums

The publication package includes the exact branch-exhaustion source, replay logs and rational certificate data. The verifier bundle contains

```
README_FINAL_BRANCH_EXHAUSTION.md
dual_components.py
verify_three_dual_rectangles_exact.py
verify_three_dual_coverage.py
verify_numerical_sanity.py
p120_slack_localization_rational_certificate.json
exact_replay.txt
coverage_replay.txt
numerical_sanity_replay.txt.
```

The bundle's own SHA-256 manifest was checked before this manuscript build; every listed file returned `OK`. The exact coverage and dense numerical scripts were also rerun in the present build environment and reproduced their terminal PASS statuses. The full exact symbolic rectangle replay is retained verbatim in the package together with its generating script; because symbolic determinant/Bernstein expansion is computationally heavier, the publication claim rests on the archived exact replay plus the independently checkable source and hashes, not on timing of a particular interactive session.

A useful reproducibility principle follows. The paper should not cite a numerical optimizer as evidence for positive definiteness when an exact replay exists. Conversely, an exact replay log should not be treated as self-authenticating: the reader should hash the source bundle, inspect the construction of the matrix being certified, and verify the feasible-set meaning of every added payment. The combination of exact arithmetic and mathematical admissibility is the proof object.

K Exact pole–pentagon mass normalization and equality geometry

The pole–pentagon lower construction is the equality geometry that fixes the candidate coefficient before the global upper-bound certificate is assembled. This appendix records the exact algebraic data used in that construction and separates the normalization calculation from the later common-measure upper-bound machinery.

Exact weights

Let the axial atom have mass p_* and let each of the five equally spaced rim atoms have mass r_*. The exact masses in the certificate archive are

$$p_* = \frac{91 - 25\sqrt{7}}{126}, \qquad r_* = \frac{7 + 5\sqrt{7}}{126}. \tag{72}$$

Their normalization is exact:

$$p_* + 5r_* = \frac{91 - 25\sqrt{7} + 35 + 25\sqrt{7}}{126} \tag{73}$$

$$= \frac{126}{126} = 1. \tag{74}$$

Thus the irrationality of the individual masses is not a numerical artifact; it cancels in the total probability law. Numerically,

$$p_* \approx 0.19733, \qquad r_* \approx 0.16053,$$

so the rim carries most of the mass while the pole remains a substantial axial component.

Five-ray angular lock

Write the five rim azimuths as

$$\theta_j = \theta_0 + \frac{2\pi j}{5}, \qquad j = 0, \dots, 4.$$

The fifth Fourier phase is therefore common:

$$e^{5i\theta_j} = e^{5i\theta_0}.$$

This is exactly the equality condition detected by the fifth-harmonic contact defect. In the degree-six majorant the sharp positive coefficient

$$\kappa_5 = \frac{-11944 + 5355\sqrt{5}}{17800} = 0.001693484242212\ldots > 0 \tag{75}$$

forces

$$D^{(6)}_{\text{contact}} \geq \frac{\kappa_5}{16}\left(R_0^2 - |R_5|^2\right).$$

Hence equality requires $|R_5| = R_0$, which is the common five-ray phase condition above. The radial double-lock then rules out unequal latitude splitting inside that angular equality class, leaving the pole plus one regular pentagonal rim.

Evaluation of the extremal coefficient

Substitution of the exact equality law into the one-sided isotropic energy functional gives

$$\mathcal{E}(\mu_*) = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9} = \alpha_3. \tag{76}$$

This symbolic evaluation is the lower-bound half of the capstone: $\Gamma_3 \geq \alpha_3$. The remainder of the proof does not optimize within the pole–pentagon family again; instead it proves that every admissible law outside its equality kernel has sufficient defect to keep the energy below the same value. This separation between an explicit lower construction and a support-independent upper certificate is useful when auditing the proof chain.

L Exact rational coverage proof for the final dual atlas

The final branch-exhaustion certificate is only global if the three rational rectangles cover the complete parameter image left by all earlier sectors. Because the annulus is parameterized by p while the dual pencil is parameterized by q, this step is best written in exact rational arithmetic.

Transforming the annulus

The remaining pole-mass interval is

$$\frac{139}{1250} \leq p \leq \frac{257}{2000}, \qquad q^2 = \frac{1 - 3p}{2}.$$

Since $q > 0$ and the map $p \mapsto q^2$ is decreasing, its exact image is

$$\frac{1229}{4000} \leq q^2 \leq \frac{833}{2500}. \tag{77}$$

The three rational chart intervals are

$$\begin{aligned} C: &\quad \frac{5543}{10000} \leq q \leq \frac{5671}{10000}, \\ B: &\quad \frac{5671}{10000} \leq q \leq \frac{5736}{10000}, \\ A: &\quad \frac{5716}{10000} \leq q \leq \frac{5773}{10000}. \end{aligned}$$

The lower endpoint of chart C lies strictly below the required square-root endpoint because

$$\frac{1229}{4000} - \left(\frac{5543}{10000}\right)^2 = \frac{30{,}725{,}000 - 30{,}724{,}849}{10^8} \tag{78}$$

$$= \frac{151}{10^8} > 0. \tag{79}$$

Similarly the upper endpoint of chart A extends beyond the required endpoint:

$$\left(\frac{5773}{10000}\right)^2 - \frac{833}{2500} = \frac{33{,}327{,}529 - 33{,}320{,}000}{10^8} \tag{80}$$

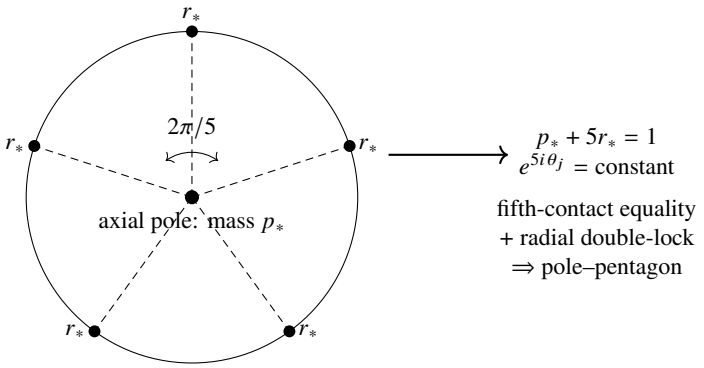


Figure 11: Exact pole–pentagon equality geometry. The angular fifth mode locks the rim to a five-ray orbit, while the radial equality conditions collapse the surviving mass to one rim latitude.

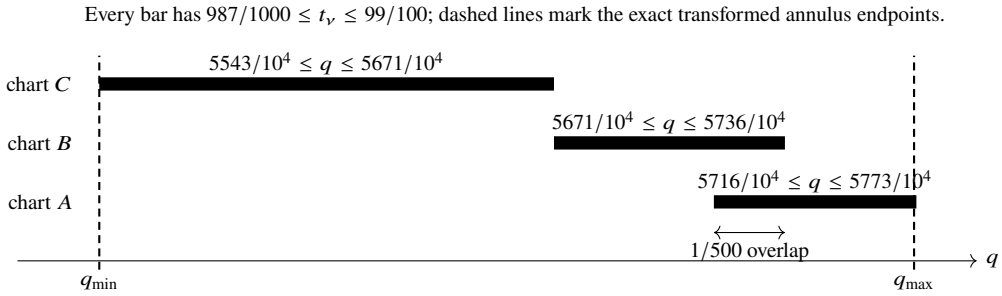


Figure 12: Exact coverage geometry. Chart C begins below the required lower endpoint, chart A extends above the upper endpoint, B meets C, and A overlaps B by $1/500$. The horizontal drawing uses the affine plotting coordinate $1000(q - 0.55)$ only for readability.

$$= \frac{7529}{10^8} > 0. \tag{81}$$

No algebraic root isolation is required: the squared endpoint comparisons are rational and have the correct orientation because every q endpoint is positive.

No internal gap

Charts C and B meet exactly at $q = 5671/10000$. Charts B and A overlap, since

$$\frac{5736}{10000} - \frac{5716}{10000} = \frac{1}{500} > 0. \tag{82}$$

All three rectangles share the same radial-correlation interval

$$\frac{987}{1000} \le t_\nu \le \frac{99}{100}.$$

Equations (79)–(82) therefore prove that the union of the three rectangles contains the entire exact q-image of the compact away-annulus. This is the arithmetic content of the released terminal status

```
PASS_EXACT_THREE_DUAL_RECTANGLES_
    COVER_FULL_AWAY_ANNULUS.
```

M The solved S_2 problem and the rank-three transition to S_3

The solved second-spectral-sum theorem is the closest finite-order precedent for the present problem, but it also shows why the S_3 proof requires genuinely new machinery. This appendix places the two arguments side by side without conflating their equality mechanisms.

Huang and Wei prove that, for every $n \ge 5$, the exact S_2 maximum is attained uniquely by a residue-dependent three-part graph K_n^*; their proof starts from Ky Fan's two-dimensional variational principle, uses edge maximality to obtain a threshold rule, forces the complement to be a chain graph, proves a weighted Ferrers spectral-tail inequality, and finally reduces the optimizer to incidence rank one. The universal coefficient $8/7$ had already been established by Kumar, Liu, Monterde, Pragada and Tait. The success of the S_2 argument therefore comes from a graph-combinatorial rigidity that remains visible after the variational reduction.

For S_3, Ky Fan variation produces a rank-three projector rather than a two-dimensional threshold geometry. The useful rigidity is no longer a chain graph in the original vertex coordinates. It is instead the geometry of a common isotropic measure in $\mathbb{R}^3$, together with the fact that all Fourier and radial moments arise from that same measure. The pole–pentagon is an equality object in this reduced geometric problem, and the hard part of

Table 7: Structural comparison between the solved S_2 theorem and the present S_3 proof architecture.

Feature	S_2	S_3
Variational object	orthonormal pair / rank-two projector	orthonormal triple / rank-three projector
Dominant reduced geometry	thresholded vertex pairs and chain-graph complement	isotropic law in $\mathbb{R}^3$ with common radial/angular carrier
Sharp coefficient	$8/7$ for the universal linear term	$\alpha_3 = \frac{1}{2} + \frac{\sqrt{5}}{18} + \frac{\sqrt{35}}{9}$
Equality model	residue-dependent three-part graph K_n^*	pole plus regular five-ray rim in the asymptotic isotropic problem
Main rigidity step	Ferrers tail inequality forces incidence rank one	rank-six, radial, fifth-contact and common-carrier defects force pole–pentagon equality
Delicate boundary	equality in the threshold rule	semidefinite endpoint charts and $t_\nu \uparrow 1$ common-carrier degeneration
Final compact step	exact integer optimization of (p, q)	support-independent rational three-rectangle dual atlas
Machine-readable role	finite exact algebra supporting a combinatorial proof	rational/interval positivity is part of the final proof object
Finite-order equality	completely classified for every $n \geq 5$	universal upper bound and asymptotic coefficient closed; full residue-dependent $M_3(n)$ classification remains open

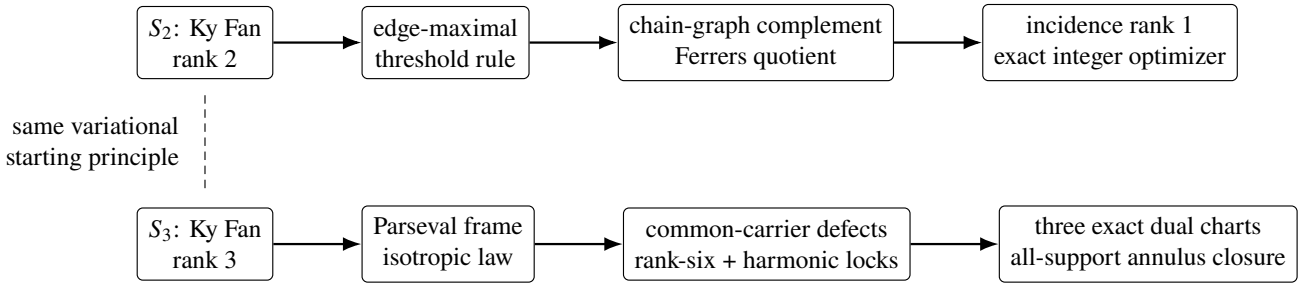


Figure 13: The methodological transition from the solved S_2 problem to S_3. Both begin with Ky Fan variation, but the successful rigidity mechanism changes from chain-graph/Ferrers structure to common-measure harmonic and semidefinite structure.

the upper bound is to preserve common-carrier coupling while excluding every other support/contact realization.

The comparison also clarifies the claim boundary of the present title. The universal S_3 extremal coefficient and the upper bound $M_3(n) \leq \alpha_3 n - 3$ are the resolved capstone of the certificate system. A finite-order residue table and an all-n uniqueness theorem analogous to Huang–Wei remain a separate second-stage problem. This distinction is mathematically substantive rather than stylistic: irrational equality masses make the passage from the asymptotic pole–pentagon law to exact integer blow-ups nontrivial.

N Further consequences and open finite-order questions

The universal coefficient theorem changes the remaining research programme. The principal unresolved questions are now refinements rather than missing sectors of the bound.

Finite-order sharpness

The inequality $M_3(n) \leq \alpha_3 n - 3$ need not be an equality at a fixed order. The pole–pentagon law has irrational weights involving $\sqrt{7}$, so finite blow-ups must approximate the ideal masses by integers. A natural next problem is therefore to determine the exact residue-dependent integer optimization near

$$np_*, \qquad nr_*, \qquad p_* + 5r_* = 1,$$

and to decide whether a finite family of rounded pole–pentagon blow-ups gives the exact $M_3(n)$ for all sufficiently large n. The solved S_2 theorem provides a precedent: the universal coefficient is only part of the story, and exact finite maxima require a separate integer/equality analysis.

Stability of near-extremizers

The proof contains several strict gaps away from the pole–pentagon kernel: low-overlap reserve, terminal harmonic reserve, fifth-contact reserve, radial mismatch reserve, interval-Hessian positivity and strict final-atlas negativity. In principle these can be assembled into a stability theorem of the form

$$\alpha_3 - \mathcal{E}(\mu) \geq C \operatorname{dist}(\mu, \mathcal{P}_5)^\gamma$$

for an appropriate metric and orbit class $\mathcal{P}_5$ of pole–pentagon measures. Such a result would identify not merely the maximal coefficient but the geometry of all near-extremal graphs after projection/blow-up.

Higher partial sums

For S_k the projection dimension becomes k, while the angular/contact geometry lives in a correspondingly larger spherical moment cone. The present S_3 analysis suggests three principles likely to survive: preserve the common measure when bounding different harmonic modes; separate equality faces from compact interiors before rational reconstruction; and prefer support-independent dual certificates when a KKT classification begins to proliferate contact topologies. Whether there is a higher-dimensional analogue of the pole–pentagon with a similarly small exact algebraic coefficient is open.

References

[1] Ky Fan. "On a theorem of Weyl concerning eigenvalues of linear transformations I". In: *Proceedings of the National Academy of Sciences USA* 35 (1949), pp. 652–655.

[2] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. 2nd ed. Cambridge University Press, 2013.

[3] Rajendra Bhatia. *Matrix Analysis*. Springer, 1997.

[4] Bojan Mohar. "On the sum of k largest eigenvalues of graphs and symmetric matrices". In: *Journal of Combinatorial Theory, Series B* 99 (2009), pp. 306–313.

[5] Kinkar Chandra Das, Seyed Ahmad Mojallal, and Shuxian Sun. "On the sum of the k largest eigenvalues of graphs and maximal energy of bipartite graphs". In: *Linear Algebra and its Applications* 569 (2019), pp. 175–194.

[6] Shaowei Sun, Yaping Min, and Kinkar Chandra Das. *Sum of the k largest eigenvalues of symmetric matrices: Theory and applications*. 2026. arXiv: 2605.26707.

[7] Ivan Gutman. "The energy of a graph". In: *Ber. Math. Stat. Sekt. Forschungsz. Graz* 103 (1978), pp. 1–22.

[8] Vladimir Nikiforov. "Beyond graph energy: Norms of graphs and matrices". In: *Linear Algebra and its Applications* 506 (2016), pp. 82–138.

[9] Vladimir Nikiforov. "Linear combinations of graph eigenvalues". In: *Electronic Journal of Linear Algebra* 15 (2006), pp. 329–336.

[10] J. Ebrahimi B., Bojan Mohar, Vladimir Nikiforov, and A. S. Ahmady. "On the sum of two largest eigenvalues of a symmetric matrix". In: *Linear Algebra and its Applications* 429 (2008), pp. 2781–2787.

[11] Mustapha Aouchiche and Pierre Hansen. "A survey of automated conjectures in spectral graph theory". In: *Linear Algebra and its Applications* 432 (2010), pp. 2293–2322.

[12] Hitesh Kumar, Lele Liu, Hermie Monterde, Shivaramakrishna Pragada, and Michael Tait. *Maximum spectral sum of graphs*. 2026. arXiv: 2604.00512.

[13] Jing Huang and Wei Wei. *The Exact Maximum of the Spectral Sum of Graphs*. 2026. arXiv: 2607.23081.

[14] Nicholas J. Higham. *Accuracy and Stability of Numerical Algorithms*. 2nd ed. SIAM, 2002.

[15] Lloyd N. Trefethen and David III Bau. *Numerical Linear Algebra*. SIAM, 1997.

[16] Saugata Basu, Richard Pollack, and Marie-Françoise Roy. *Algorithms in Real Algebraic Geometry*. 2nd ed. Springer, 2006.

[17] Philippe Delsarte, Jean-Marie Goethals, and Johan J. Seidel. "Spherical codes and designs". In: *Geometriae Dedicata* 6 (1977), pp. 363–388.

[18] I. J. Schoenberg. "Positive definite functions on spheres". In: *Duke Mathematical Journal* 9 (1942), pp. 96–108.

[19] John J. Benedetto and Matthew Fickus. "Finite normalized tight frames". In: *Advances in Computational Mathematics* 18 (2003), pp. 357–385.

[20] Peter G. Casazza and Gitta Kutyniok, eds. *Finite Frames: Theory and Applications*. Birkhäuser, 2013.

[21] P. W. H. Lemmens and J. J. Seidel. "Equiangular lines". In: *Journal of Algebra* 24 (1973), pp. 494–512.

[22] John H. Conway, Ronald H. Hardin, and Neil J. A. Sloane. "Packing lines, planes, etc.: Packings in Grassmannian spaces". In: *Experimental Mathematics* 5 (1996), pp. 139–159.

[23] Matthew Fickus, John Jasper, and Dustin G. Mixon. "Packings in real projective spaces". In: *SIAM Journal on Applied Algebra and Geometry* 2 (2018), pp. 377–409.

[24] Brendan D. McKay. "Practical graph isomorphism". In: *Congressus Numerantium* 30 (1981), pp. 45–87.

[25] Brendan D. McKay and Adolfo Piperno. "Practical graph isomorphism, II". In: *Journal of Symbolic Computation* 60 (2014), pp. 94–112.

[26] Ronald C. Read. "Every one a winner or how to avoid isomorphism search when cataloguing combinatorial configurations". In: *Annals of Discrete Mathematics* 2 (1978), pp. 107–120.

[27] I. A. Faradžev. "Constructive enumeration of combinatorial objects". In: *Problèmes combinatoires et théorie des graphes*. 1978, pp. 131–135.

[28] Ulf Grenander and Gábor Szegő. *Toeplitz Forms and Their Applications*. University of California Press, 1958.

[29] Lipót Fejér. "Über trigonometrische Polynome". In: *Journal für die reine und angewandte Mathematik* (1915).

[30] Frigyes Riesz. "Über ein Problem des Herrn Carathéodory". In: *Journal für die reine und angewandte Mathematik* (1916).

[31] Mihai Putinar. “Positive polynomials on compact semi-algebraic sets”. In: *Indiana University Mathematics Journal* 42 (1993), pp. 969–984.

[32] Jean B. Lasserre. “Global optimization with polynomials and the problem of moments”. In: *SIAM Journal on Optimization* 11 (2001), pp. 796–817.

[33] Pablo A. Parrilo. “Semidefinite programming relaxations for semialgebraic problems”. In: *Mathematical Programming* 96 (2003), pp. 293–320.

[34] Monique Laurent. “Sums of squares, moment matrices and optimization over polynomials”. In: *Emerging Applications of Algebraic Geometry*. Vol. 149. IMA Volumes in Mathematics and its Applications. Springer, 2009, pp. 157–270.

[35] Christine Bachoc and Frank Vallentin. “New upper bounds for kissing numbers from semidefinite programming”. In: *Journal of the American Mathematical Society* 21 (2008), pp. 909–924.

[36] David de Laat and Frank Vallentin. “A semidefinite programming hierarchy for packing problems in discrete geometry”. In: *Mathematical Programming* 151 (2015), pp. 529–553.

[37] Chandler Davis and W. M. Kahan. “The rotation of eigenvectors by a perturbation. III”. In: *SIAM Journal on Numerical Analysis* 7 (1970), pp. 1–46.